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Advanced Technological Institute – Gampaha
Higher National Diploma in Information Technology

2nd Year Individual Project

Final Report



**Development of a Web-Based Learning Management System (LMS) For
Aura Pen Saku Education Institute**

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DECLARATION

I hereby declare that this project report, titled "**Development of a Web-Based Learning Management System (LMS) for Aura Pen Saku Education Institute,**" is my own original work carried out under the supervision of **Ms. N.D. Gunasekara**.

This report is submitted to fulfill the requirements of the Higher National Diploma in Information Technology (HNDIT) at the Sri Lanka Institute of Advanced Technological Education (SLIATE).

This project work has not been submitted previously, in whole or in part, for any degree, diploma, or other academic award at any other institute or university. All external sources of information used in this report have been duly acknowledged and referenced in accordance with the IEEE referencing style.

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CHAPTER 1 – INTRODUCTION

1.1 Organization Details

Aura Pen Saku Education Institute is a tuition institute located opposite Bandaranayaka College, Mirigama, that provides supplementary academic classes to school students. The institute is led by Mr. Sameera Kaluarachchi, who acted as the client for this project.

1.1.1 Organizational Structure

The institute is organized around an administrator who oversees the overall operation, a group of subject teachers who are responsible for preparing and delivering lesson content, and the students who are grouped by grade and subject. Prior to this project, all of these roles were coordinated manually, with the administrator and teachers keeping physical records of students, class schedules, and learning materials.

1.2 Motivation for the Project

1.2.1 Existing System

Before this system was developed, Aura Pen Saku Education Institute did not have any digital platform. Student registrations, subject enrolments, class timetables, and learning materials such as tutes and notes were all managed manually using paper records and informal messaging platforms.

1.2.2 Current Issues

- Student and teacher records were kept manually, making them difficult to search, update, and back up.
- There was no systematic, grade-wise or subject-wise categorization of lesson content, so tutes and notes were distributed inconsistently.
- Assessing student understanding relied on paper-based tests that had to be marked manually, delaying feedback.
- There was no central place for students to check their timetable, download materials, or track their progress.
- Communication between students, teachers, and administration relied on informal, unstructured channels.

1.3 Project Aims and Objectives

1.3.1 Aim

The aim of this project is to design and develop a comprehensive, web-based Learning Management System (LMS) that automates educational content distribution, assessment, and student management for Aura Pen Saku Education Institute, replacing the institute's manual, paper-based processes.

1.3.2 Specific Objectives

- To design and develop a secure, role-based user authentication system for Admins, Teachers, and Students.
- To implement a grade-wise and subject-wise content management system that allows teachers to upload learning materials.
- To provide a personalized dashboard for students displaying relevant resources, timetables, and notices.
- To develop an online assessment module including automated marking quizzes.
- To build a centralized MySQL database that reliably stores institutional and academic data.

1.4 Problem Definition

Aura Pen Saku Education Institute currently manages all its academic records, lesson content, and assessments manually. There is no systematic digital process through which learning materials can be categorized and shared according to grade and subject. This project addresses the problem by replacing the manual workflow with a role-based web application.

1.5 Scope of the System

The system covers user authentication for three roles (Admin, Teacher, Student); management of students, teachers, and subjects; grade-wise learning material management (PDF tutes and video links); timetable management; online quiz module with automated marking; student progress tracking; and dynamic announcements. Online payments, live video streaming, and mobile apps are outside the scope.

1.6 Organization of the Dissertation

This report is organized into six chapters. Chapter 1 introduces the project. Chapter 2 presents system analysis. Chapter 3 covers system design. Chapter 4 describes system implementation. Chapter 5 presents evaluation and testing. Chapter 6 concludes with future enhancements.

CHAPTER 2 – SYSTEM ANALYSIS

2.1 Requirement Gathering Methods

To ensure that the **Aura Pen Saku LMS** meets all functional and operational expectations of the institute, a comprehensive requirement gathering process was conducted. Both primary and secondary data collection techniques were utilized to gather precise information from stakeholders:

2.1.1 Stakeholder Interviews & Client Discussions:

- Direct structured and semi-structured interview sessions were conducted with the principal client, **Mr. Sameera Kaluarachchi**, and administrative staff.
- Key topics discussed included overall institute administration workflows, student enrollment procedures, fee structures, class scheduling challenges, and required security levels for different user roles (Admin, Teacher, Student).

2.1.2 Reviewing Existing Documents & Manual Records:

- Physical documents currently used by the institute were thoroughly analyzed.
- This included physical attendance registers, paper tutes/notes, hand-written timetables, physical mark sheets, and wall-mounted noticeboard slips.
- Analyzing these documents helped identify the exact data fields required for database entities (e.g., students, teachers, subjects, student_results).

2.1.3 Direct Observation of Operations:

- Daily operational activities at the institute were observed firsthand during class hours.
- Special attention was given to how paper tutes are handed out, how whiteboard class schedule updates are communicated, and how teachers grade MCQ test sheets by hand.

2.2 Workflow of the Current System

The existing operational workflow at the institute relies entirely on traditional, paper-based manual processes. The step-by-step current manual workflows and their associated operational drawbacks are detailed in Table 2.1 below:

Operational Activity	Current Manual Workflow	Drawbacks / Operational Limitations
Study Material & Tute Distribution	Teachers print physical tute booklets and paper notes at a print shop, which are handed out to students in person during class.	High recurring paper and printing costs; absent students cannot access learning materials.
Quizzes & Exam Evaluation	Exams are printed on paper, conducted in class, and graded manually by hand before writing scores in physical mark sheets.	Extremely time-consuming for teachers; delay of days in releasing results; risk of human error during manual grading.
Class Schedule & Timetable Management	Schedules are written on classroom whiteboards or printed and posted on physical noticeboards at the institute entrance.	Inability to communicate last-minute class changes in real-time; students frequently miss rescheduled classes.
Notice & Announcement Publishing	Official notices (exam dates, holiday notices, fee reminders) are typed, printed, and pinned onto wall noticeboards.	Students who do not physically visit the institute on that day miss important and urgent announcements.
Record Keeping & Data Storage	All student profiles, enrollment lists, attendance records, and exam scores are kept in physical paper registers and logbooks.	Physical files occupy large space, suffer from physical decay/damage, and risk permanent data loss with zero backup.

Table 2-1 : Workflow of system

2.3 System Requirement Specification (SRS)

The System Requirement Specification defines the functional and non-functional requirements of the proposed AuraPen LMS web application.

2.2.1 Functional Requirements

Functional requirements describe the specific actions and features that the system performs for each user role.

Module 01: Authentication & Security

- The system shall allow users to log in using a valid username and password.
- The system shall validate user credentials against the database and redirect them to their specific dashboard (Admin, Teacher, or Student).
- The system shall restrict unauthorized users from accessing protected pages
- The system shall allow users to securely log out and end their active session.

Module 02: Admin Dashboard

- The system shall allow the Admin to add, edit, and delete teacher records.
- The system shall allow the Admin to register, update, and remove student records with assigned grades and subjects.
- The system shall allow the Admin to add and manage subject master records.
- The system shall allow the Admin to create, update, and delete weekly class schedules on the timetable.

Module 03: Teacher Dashboard

- The system shall allow teachers to upload PDF notes, tutes, and video links filtered by grade and subject.
- The system shall allow teachers to create multiple-choice question (MCQ) quizzes with time limits.
- The system shall allow teachers to manage a reusable question bank for exams.
- The system shall allow teachers to view the list of students enrolled in their classes and publish announcements.

Module 04: Student Dashboard

- The system shall display a personalized dashboard with the student's registered subjects and timetable.
- The system shall display live institute notices fetched directly from the database.
- The system shall allow students to view and download PDF study materials for their grade.
- The system shall allow students to attempt online quizzes and view instant auto-graded results.
- The system shall allow students to submit feedback and view admin replies.

2.2.2 Non-Functional Requirements

Non-functional requirements describe the quality standards, performance metrics, security levels, and operational constraints of the LMS.

- **Performance & Response Time:** All web pages, API queries, and file downloads must load within 2 to 3 seconds under normal network conditions.
- **Security & Data Protection:** All user passwords must be securely hashed before storing in MySQL. Role-based routing must block unauthorized URL access.
- **Usability & UX:** The interface must be clean and responsive, allowing users to reach any learning material within three clicks.
- **Reliability & Integrity:** Normalized primary and foreign keys must maintain data integrity across database tables. Fallback data handling must prevent UI crashes.
- **Scalability:** The architecture must easily handle an increasing volume of registered students and uploaded study materials without loss of performance.

2.4 Feasibility Study

A feasibility study was conducted to evaluate the technical, economic, and operational practicality of developing the web-based LMS for Aura Pen Saku Education Institute.

2.4.1 Technical Feasibility

The project is technically feasible as it uses reliable, industry-standard modern web technologies:

- **Frontend:** React.js for building a fast, interactive single-page user interface.
- **Backend:** Node.js with Express.js to handle secure REST API connections.
- **Database:** MySQL 8.0 (XAMPP) for structured relational database management.
- **Hardware Requirements:** Standard computer hardware (Core i5, 16GB RAM) was fully sufficient for local development and testing without high-end server hardware.

2.4.2 Economic Feasibility

- The system is 100% economically feasible with zero software licensing costs:
- All development platforms (React.js, Node.js, Express.js, MySQL, VS Code) are free and open-source.
- The only future cost for the client will be standard web hosting and domain registration for live deployment.

2.4.3 Operational Feasibility

The system is operationally feasible as it simplifies the institute's daily activities:

- **Admin:** Manages user accounts and timetable matrix easily using dynamic table interfaces.
- **Teachers:** Directly upload PDF handouts and lesson links instead of printing paper tutes.
- **Students:** Access study notes, timetables, and notices online via Chrome or Edge browsers without installing external software.

2.5 Assumptions & Limitations

2.5.1 Assumptions

- **Accurate Content:** Teachers will upload accurate PDF notes, correct lesson links, and valid quiz answer keys.
- **Device Access:** Administrators, teachers, and students have access to a PC, laptop, or smartphone with internet connectivity.
- **Standard Web Browsers:** Users will access the portal using updated modern browsers such as Google Chrome or Microsoft Edge.

2.5.2 Limitations

- **No Online Payments:** Student fee processing, payment gateways, and invoice management are outside the scope of this project.
- **No Live Streaming:** Built-in live video conferencing (like Zoom) is not included; teachers share YouTube video links instead.
- **No Native Mobile Apps:** Standalone Android (.apk) or iOS (.ipa) mobile apps are not provided, though the web app is mobile-responsive.

CHAPTER 3 – SYSTEM DESIGN

3.1 Use Case Diagram

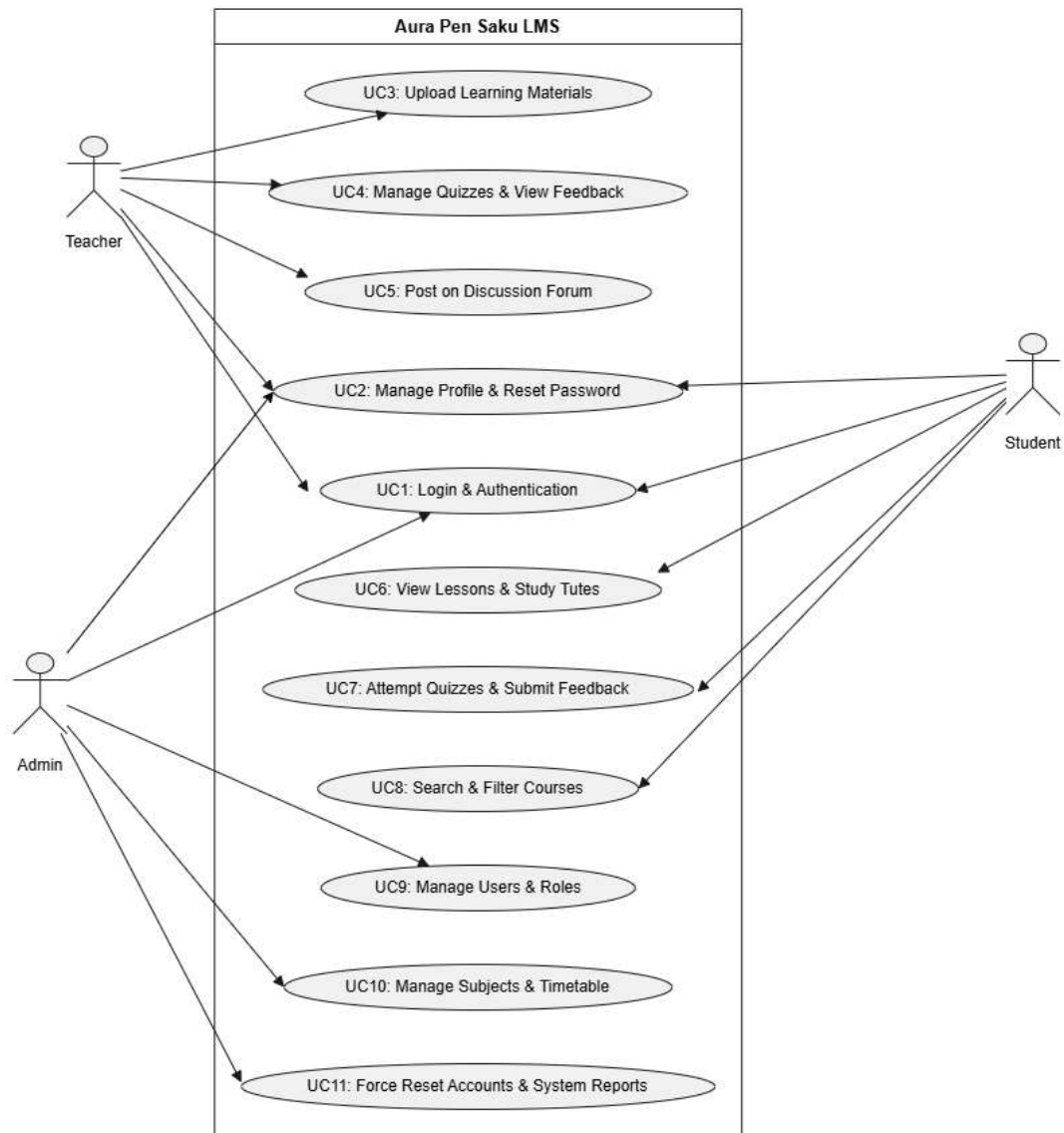


Figure 3.1 Use case Diagram

3.1.1 Use Case Scenario: UC1 - Login & Authentication

Field	Details
Use Case Name	UC1: Login & Authentication
Actors	Admin, Teacher, Student
Description	Authenticates users against the database and redirects them to their respective role-based dashboards.
Preconditions	User account exists in the users table with active status.
Postconditions	User session is created and redirected to dashboard.
Basic Flow	1. User enters username and password. 2. System validates credentials. 3. System redirects user to role dashboard.
Exceptions	Invalid credentials display <i>"Invalid username or password"</i> .

Table 3.1 use case 01: Login

3.1.2 Use Case Scenario: UC2 - Manage Profile & Reset Password

Field	Details
Use Case Name	UC2: Manage Profile & Reset Password
Actors	Admin, Teacher, Student
Description	Allows users to update contact details and reset passwords (including first-time login forced reset).
Preconditions	User is logged into the system.
Postconditions	Profile or password updated in the database.
Basic Flow	1. User enters new profile details/password. 2. System validates input. 3. System updates database and displays success notice.
Exceptions	Invalid input format prompts error alert.

Table 3- 2 Use case 02:Manage Profile

3.1.3 Use Case Scenario: UC3 - Upload Learning Materials

Field	Details
Use Case Name	UC3: Upload Learning Materials
Actors	Teacher
Description	Enables teachers to upload PDF study notes and tutes for their subjects.
Preconditions	Teacher is logged in and assigned to the subject.
Postconditions	File saved on server; file path stored in learning_materials table.
Basic Flow	1. Teacher selects subject and attaches PDF file. 2. System checks file format and size. 3. System uploads file and displays under study materials.
Exceptions	Invalid file format (.exe/.mp3) or file size > 50MB is rejected.

Table 3-3 Use case 03 Learning Materials

3.1.4 Use Case Scenario: UC4 - Manage Quizzes & View Feedback

Field	Details
Use Case Name	UC4: Manage Quizzes & View Feedback
Actors	Teacher
Description	Allows teachers to create online MCQ quizzes, set time limits, and review student feedback.
Preconditions	Teacher is logged in.
Postconditions	Quiz saved in quizzes table and questions saved in questions table.
Basic Flow	1. Teacher sets quiz title and timer. 2. Teacher inputs MCQ questions and correct options. 3. System publishes quiz for students.
Exceptions	Missing correct answer selection prompts validation error.

Table 3- 4 Use case 04 Manage Quiz

3.1.5 Use Case Scenario: UC5 - Post on Discussion Forum

Field	Details
Use Case Name	UC5: Post on Discussion Forum
Actors	Teacher
Description	Allows teachers to post discussion topics and announcements for subject modules.
Preconditions	Teacher is logged in.
Postconditions	Post stored in database and shown on subject forum feed.
Basic Flow	1. Teacher types topic/message. 2. System records timestamp and posts message to forum board.
Exceptions	Blank message body is blocked.

Table 3-5 Use case 05 :Discussion forum

3.1.6 Use Case Scenario: UC6 - View Lessons & Study Tutes

Field	Details
Use Case Name	UC6: View Lessons & Study Tutes
Actors	Student
Description	Allows students to view lesson topics and download PDF tutes uploaded by teachers.
Preconditions	Student is logged in and enrolled in the subject.
Postconditions	PDF file downloaded or viewed on screen.
Basic Flow	1. Student selects enrolled subject. 2. Student clicks 'Download PDF'. 3. System serves PDF file stream.
Exceptions	Missing file path shows " <i>Material temporarily unavailable</i> ".

Table 3-6 Use case 06 :View lesson

3.1.7 Use Case Scenario: UC7 - Attempt Quizzes & Submit Feedback

Field	Details
Use Case Name	UC7: Attempt Quizzes & Submit Feedback
Actors	Student
Description	Enables students to take timed MCQ quizzes with instant auto-grading and submit course feedback.
Preconditions	Student is enrolled in the course and quiz is open.
Postconditions	Quiz result saved in student_results table; feedback stored in feedbacks table.
Basic Flow	1. Student answers MCQ questions within time limit. 2. Student submits quiz. 3. System auto-evaluates marks and displays score.
Exceptions	Quiz timer expiration (00:00) triggers automatic submission.

Table 3-7 Use case 07 :Quizzes

3.1.8 Use Case Scenario: UC8 - Search & Filter Courses

Field	Details
Use Case Name	UC8: Search & Filter Courses
Actors	Student
Description	Allows students to search subjects by course name, code, or grade filter.
Preconditions	Student portal is active.
Postconditions	Filtered course list displayed on portal UI.
Basic Flow	1. Student types keyword or selects grade dropdown. 2. System queries subjects table. 3. System renders matching subjects.
Exceptions	No match found displays "No subjects found".

Table 3-8 Use case 08 :Search

3.1.9 Use Case Scenario: UC9 - Manage Users & Roles

Field	Details
Use Case Name	UC9: Manage Users & Roles
Actors	Admin
Description	Allows Admin to create, update, or deactivate Teacher and Student accounts.
Preconditions	Admin logged into Admin Portal.
Postconditions	Record updated in users, teachers, or students tables.
Basic Flow	1. Admin inputs user details and selects role. 2. System checks email/username uniqueness. 3. System saves user account.
Exceptions	Duplicate username/email displays " <i>User already exists</i> ".

Table 3-9 Use case 06 :Manage users

3.1.10 Use Case Scenario: UC10 - Manage Subjects & Timetable

Field	Details
Use Case Name	UC10: Manage Subjects & Timetable
Actors	Admin
Description	Enables Admin to create subjects, assign teachers, and manage the live matrix timetable.
Preconditions	Admin logged into system.
Postconditions	Record saved in subjects, teacher_subjects, or timetable tables.
Basic Flow	1. Admin inputs subject details or schedule slot. 2. System checks schedule conflicts. 3. System updates database and syncs live timetable.
Exceptions	Time conflict displays " <i>Slot already scheduled</i> ".

Table 3-10 Use case 06 :Subjects

3.1.11 Use Case Scenario: UC11 - Force Reset Accounts & System Reports

Field	Details
Use Case Name	UC11: Force Reset Accounts & System Reports
Actors	Admin
Description	Allows Admin to issue temporary password overrides and view system analytics.
Preconditions	Admin logged in with administrative privileges.
Postconditions	Password flag set to is_first_login = TRUE; system report displayed.
Basic Flow	1. Admin selects user and triggers force reset. 2. System issues temporary credentials. 3. System prompts confirmation message.
Exceptions	User account not found displays error alert.

Table 3-11 Use case 11 :View lesson

3.2 Class Diagram

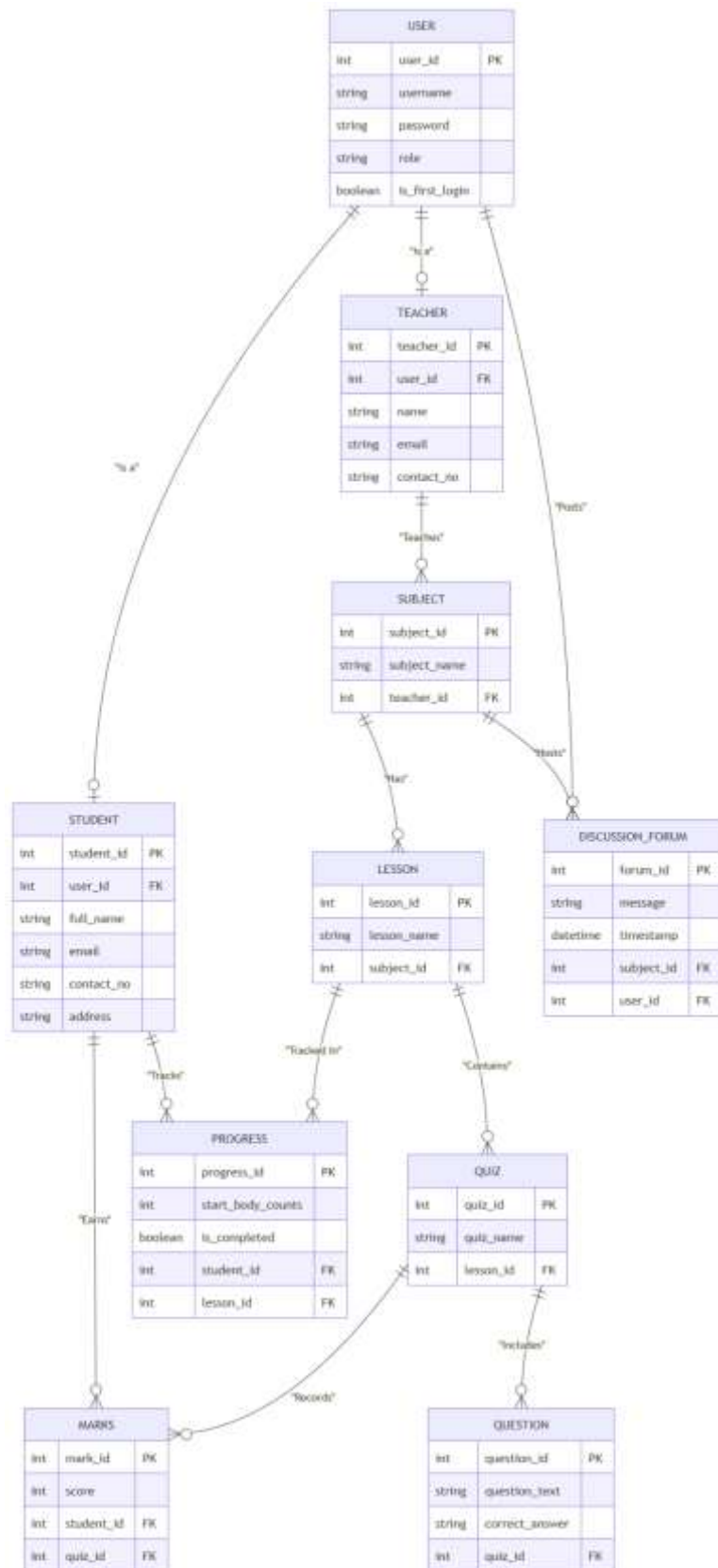


Figure 2 Class Diagram

3.3 ER Diagram

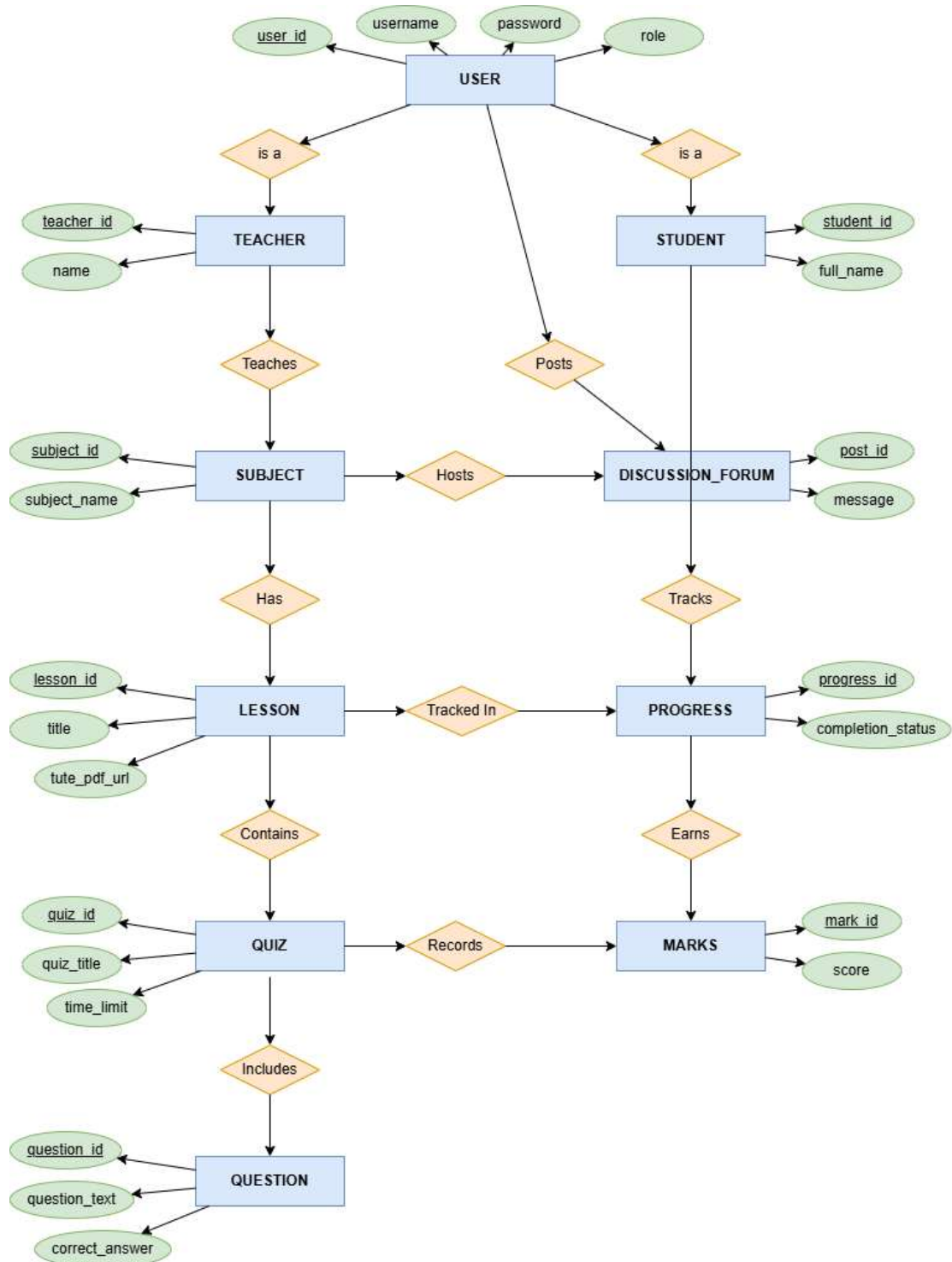


Figure 3 ER Diagram

3.4 Database Design

This section describes the logical and physical database structure for the **Aura Pen Saku LMS**.

3.4.1 Relational Schema

The Database schema is mapped from the ER Diagram into normalized relational tables (3NF):

- USER (user_id, username, password, role, is_first_login)
- TEACHER (teacher_id, user_id, name, email, contact_no)
- STUDENT (student_id, user_id, full_name, email, contact_no, address)
- SUBJECT (subject_id, subject_name, teacher_id)
- LESSON (lesson_id, lesson_name, subject_id)
- QUIZ (quiz_id, quiz_name, lesson_id)
- QUESTION (question_id, question_text, correct_answer, quiz_id)
- MARKS (mark_id, score, student_id, quiz_id)
- PROGRESS (progress_id, start_body_counts, is_completed, student_id, lesson_id)
- DISCUSSION_FORUM (forum_id, message, timestamp, subject_id, user_id)

3.4.2 Data Dictionary

Column Name	Data Type	Constraint	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for each user
username	VARCHAR(50)	UNIQUE, NOT NULL	User login username
password	VARCHAR(255)	NOT NULL	Encrypted password
role	VARCHAR(20)	NOT NULL	Role (Admin/Teacher/Student)
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Registration timestamp

Table 3. 2-2user

Table 3.3: Teacher

Column Name	Data Type	Constraint	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for teacher
user_id	INT	FOREIGN KEY (users)	Associated user account ID
full_name	VARCHAR(100)	NOT NULL	Teacher's full name
email	VARCHAR(100)	UNIQUE, NOT NULL	Contact email address
phone	VARCHAR(15)	NOT NULL	Contact phone number

*Table 3. 2-3 teacher***Table 3.4: students**

Column Name	Data Type	Constraint	Description
Id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for student
user_id	INT	FOREIGN KEY (users)	Associated user account ID
full_name	VARCHAR(100)	NOT NULL	Student's full name
email	VARCHAR(100)	UNIQUE, NOT NULL	Contact email address
phone	VARCHAR(15)	NOT NULL	Contact phone number

Table 3. 2-4 students

Table 3.5: subjects Table

Column Name	Data Type	Constraint	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for subject
subject_code	VARCHAR(20)	UNIQUE, NOT NULL	Official subject code
subject_name	VARCHAR(100)	NOT NULL	Name of the subject

*Table 3. 2-5 subject Table***Table 3.6: teacher_subjects Table**

Column Name	Data Type	Constraint	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Mapping record ID
teacher_id	INT	FOREIGN KEY (teachers)	Assigned teacher ID
subject_id	INT	FOREIGN KEY (subjects)	Assigned subject ID

Table 3. 2-6 teacher_subjects

Table 3.7: student_subjects Table

Column Name	Data Type	Constraint	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Enrollment mapping ID
student_id	INT	FOREIGN KEY (students)	Enrolled student ID
subject_id	INT	FOREIGN KEY (subjects)	Enrolled subject ID

*Table 3. 2-7 student_subjects***Table 3.8: learning_materials Table**

Column Name	Data Type	Constraint	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique material file ID
title	VARCHAR(150)	NOT NULL	Material title / note name
file_path	VARCHAR(255)	NOT NULL	Server file path for PDF/Doc
subject_id	INT	FOREIGN KEY (subjects)	Associated subject ID

Table 3. 2-8 learning Materials

Table 3.9: lesson_activities Table

Column Name	Data Type	Constraint	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique lesson activity ID
activity_title	VARCHAR(150)	NOT NULL	Activity name
subject_id	INT	FOREIGN KEY (subjects)	Associated subject ID

*Table 3. 2-9 Lesson Activities***Table 3.10: quizzes Table**

Column Name	Data Type	Constraint	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique quiz ID
quiz_title	VARCHAR(150)	NOT NULL	Title of the quiz
subject_id	INT	FOREIGN KEY (subjects)	Associated subject ID

Table 3. 2-10 Quizzes

Table 3.11: questions Table

Column Name	Data Type	Constraint	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique question ID
quiz_id	INT	FOREIGN KEY (quizzes)	Associated quiz ID
question_text	TEXT	NOT NULL	Question statement
correct_answer	VARCHAR(255)	NOT NULL	Correct option value

*Table 3. 2-11 Question Table***Table 3.12: question_bank Table**

Column Name	Data Type	Constraint	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Master question bank ID
subject_id	INT	FOREIGN KEY (subjects)	Subject category
question	TEXT	NOT NULL	Reusable question text

Table 3. 2-12 Question_Ban

Table 3.13: Student_result Table

Column Name	Data Type	Constraint	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Result record ID
student_id	INT	FOREIGN KEY (students)	Student ID
quiz_id	INT	FOREIGN KEY (quizzes)	Quiz ID
marks_obtained	INT	NOT NULL	Total score earned

*Table 3. 2-13 student_result***Table 3.14: announcements Table**

Column Name	Data Type	Constraint	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Notice/Announcement ID
title	VARCHAR(150)	NOT NULL	Announcement headline
content	TEXT	NOT NULL	Message details
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Date posted

Table 3. 2-14 announcement

Table 3.15: feedbacks Table

Column Name	Data Type	Constraint	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Feedback record ID
student_id	INT	FOREIGN KEY (students)	Student who submitted feedback
message	TEXT	NOT NULL	Feedback comment

*Table 3. 2-15 feedback***Table 3.16: timetable Table**

Column Name	Data Type	Constraint	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Schedule entry ID
subject_id	INT	FOREIGN KEY (subjects)	Scheduled subject ID
day_of_week	VARCHAR(15)	NOT NULL	Class day
start_time	TIME	NOT NULL	Class start time
end_time	TIME	NOT NULL	Class end time

Table 3. 2-16 timetable

3.4.3 Database Integrity Constraints & Rules

This section outlines the rules and database constraints applied to the **Aura Pen Saku LMS** database to ensure data accuracy, consistency, and security.

Primary Key Constraint:

- Every table in the system (e.g., users, students, subjects, quizzes) contains a unique Primary Key (id), which uniquely identifies each record and prevents duplicate entries.

Foreign Key Constraint & Referential Integrity:

- Relationships between entities are strictly enforced using Foreign Keys (e.g., user_id in teachers table referencing id in users table).
- If a parent record is deleted, associated child records are handled gracefully using CASCADE or SET NULL actions to avoid orphan records.

Entity & Value Integrity (NOT NULL):

- Critical attributes such as username, password, subject_code, and marks_obtained are configured with NOT NULL constraints to ensure required data cannot be left blank during data insertion.

Uniqueness Constraint (UNIQUE):

- Attributes like username in users table and email in students / teachers tables are assigned UNIQUE constraints to prevent duplicate user accounts or email registrations.

Domain & Data Type Constraints:

- Strict data types are applied to restrict input values (e.g., INT for scores and IDs, DATETIME for timestamps, and VARCHAR for textual inputs) ensuring only valid data formats are stored.

3.5 Interface Design

The user interface of the **Aura Pen Saku LMS** is designed using React.js and CSS3, ensuring a responsive and intuitive layout across both desktop and mobile screens.

3.5.1 Authentication & Profile Interfaces

1. System Login Interface

Allows Admin, Teacher, and Student users to authenticate using valid credentials and redirects them to role-specific dashboards.

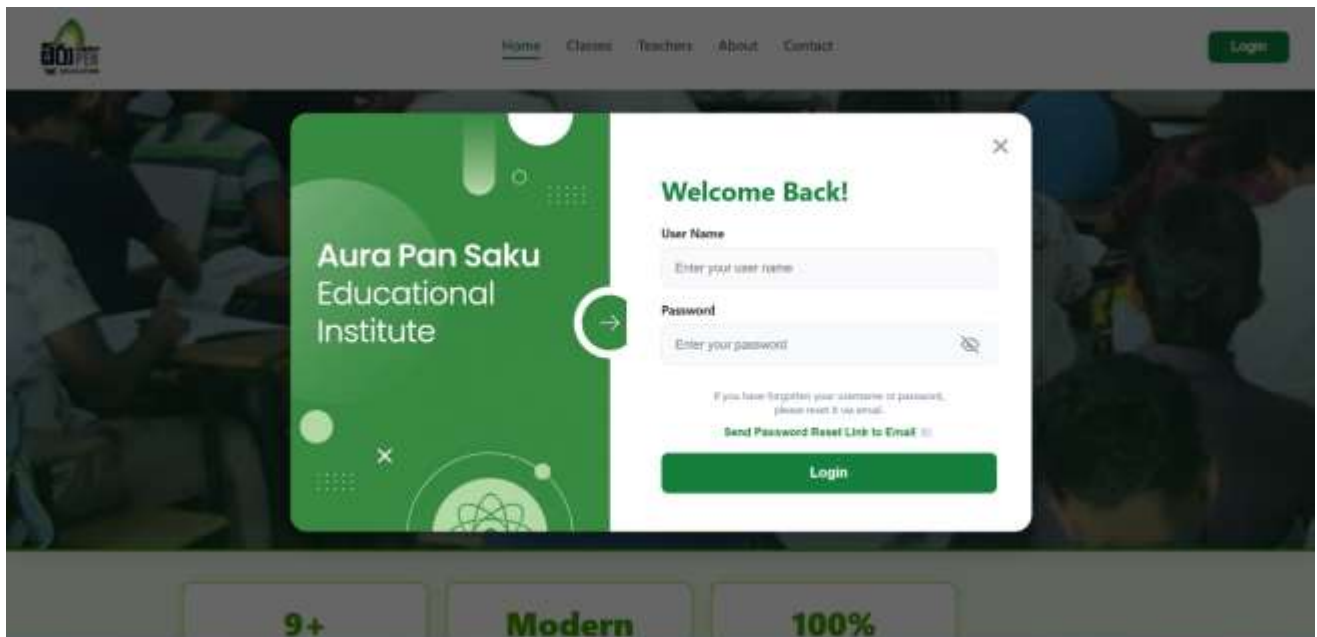


Figure 4 Login Screen

2. Institute Overview & Features Section

Highlights the institutional strengths of AuraPen Saku Education Institute, detailing expert subject guidance, model paper discussions, spacious classrooms, and an integrated digital LMS environment.



Figure 5

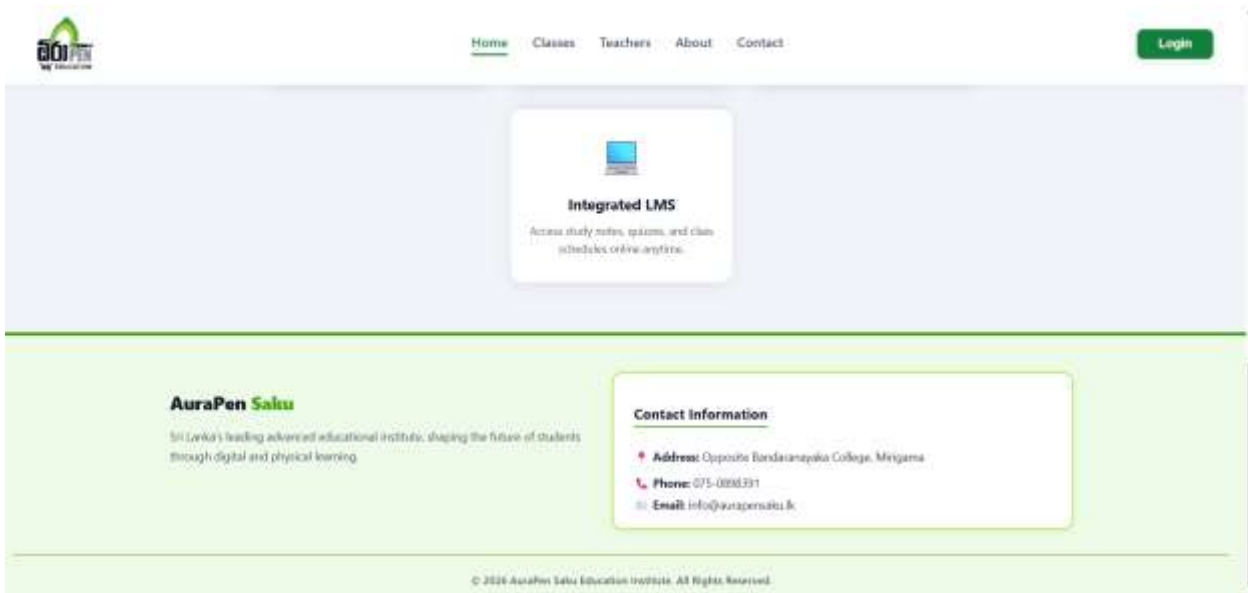


Figure 6

3. Public Live Class Timetable Portal

Displays an interactive, grade-wise filtered schedule (Grade 6 to Grade 12) for all active classes, showing assigned days, time slots, subjects, and assigned lecturers without requiring prior system login.

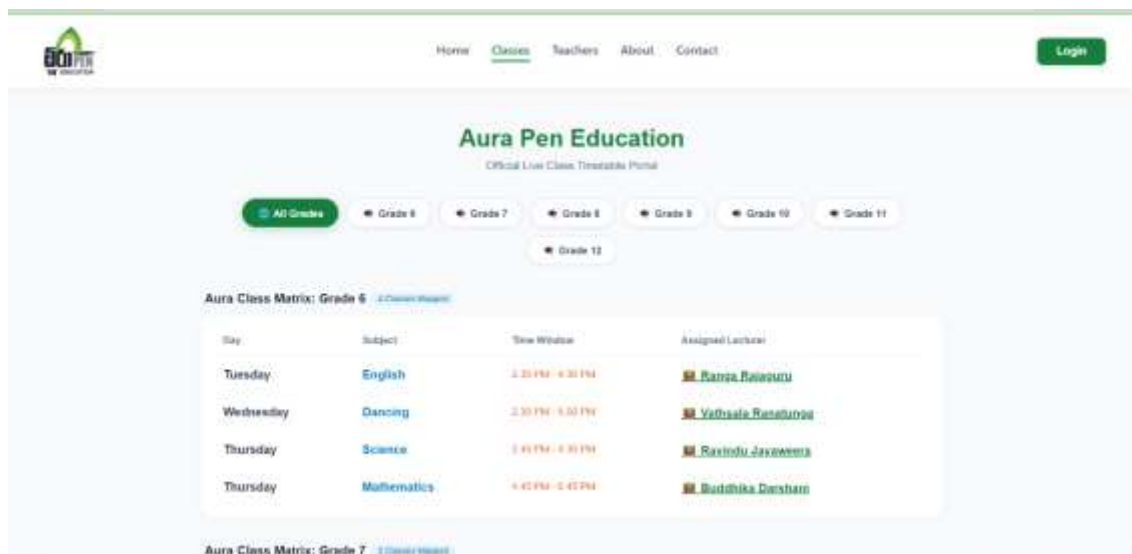


Figure 7 Public Timetable Matrix Screen

4. Institute About Us Page

Provides institutional metadata detailing the vision, mission, and educational goals of AuraPen Saku Education Institute in empowering students from secondary to Advanced Level examinations.

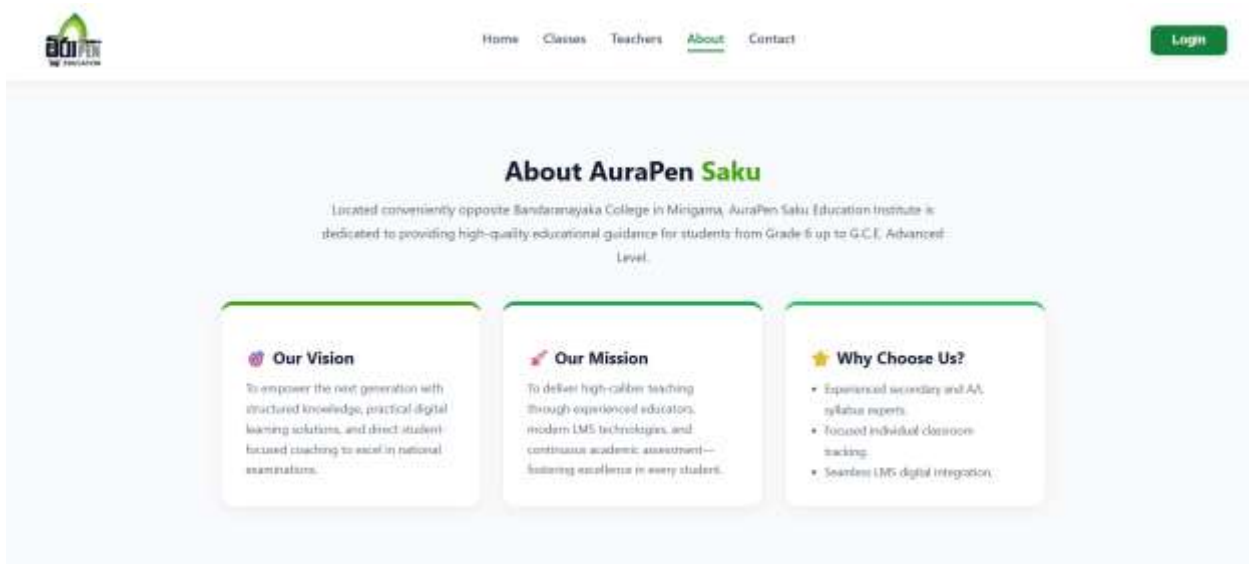


Figure 8 About Us screen

5. Public Live Class Timetable Portal

Displays an interactive, grade-wise filtered schedule (Grade 6 to Grade 12) for all active classes, showing assigned days, time slots, subjects, and assigned lecturers without requiring prior system login.

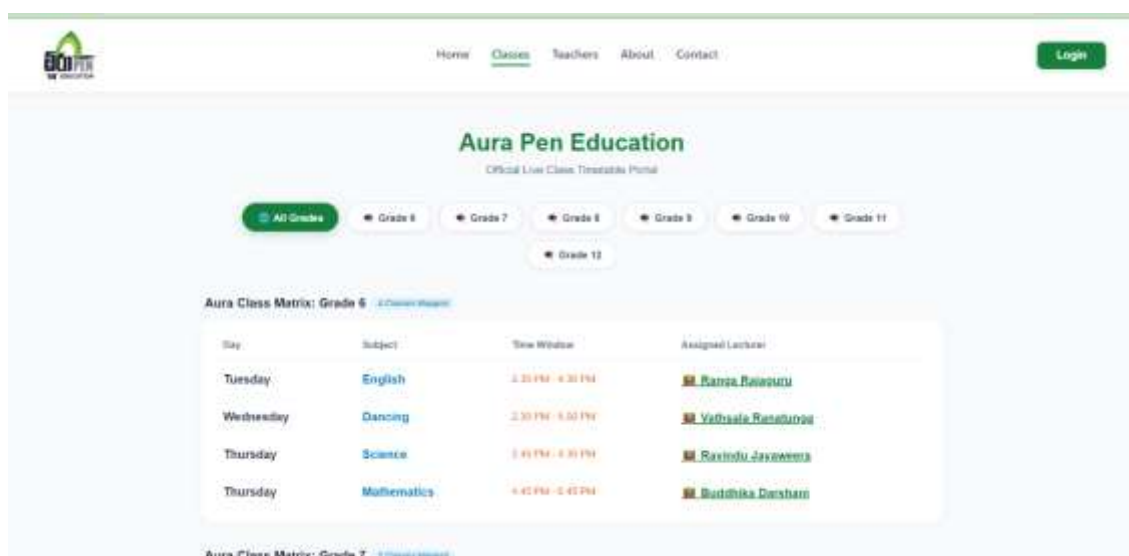


Figure 9 Public Timetable Matrix Screen

6. Faculty & Teacher Directory Interface

Presents a searchable card grid of the institute's teaching panel, highlighting each teacher's qualifications, core subjects, phone numbers, and official email addresses.

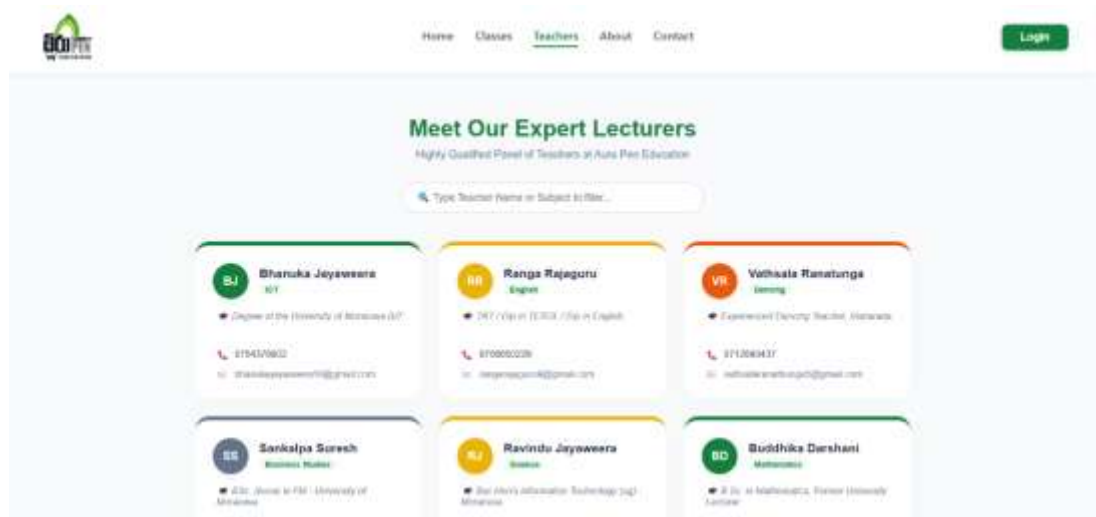


Figure 10 Lecturer Directory Screen

7. Public Contact & Inquiry Portal

Offers an direct contact channel displaying institutional location details, phone hotlines, and an interactive form for visitors to submit general inquiries directly to the management.

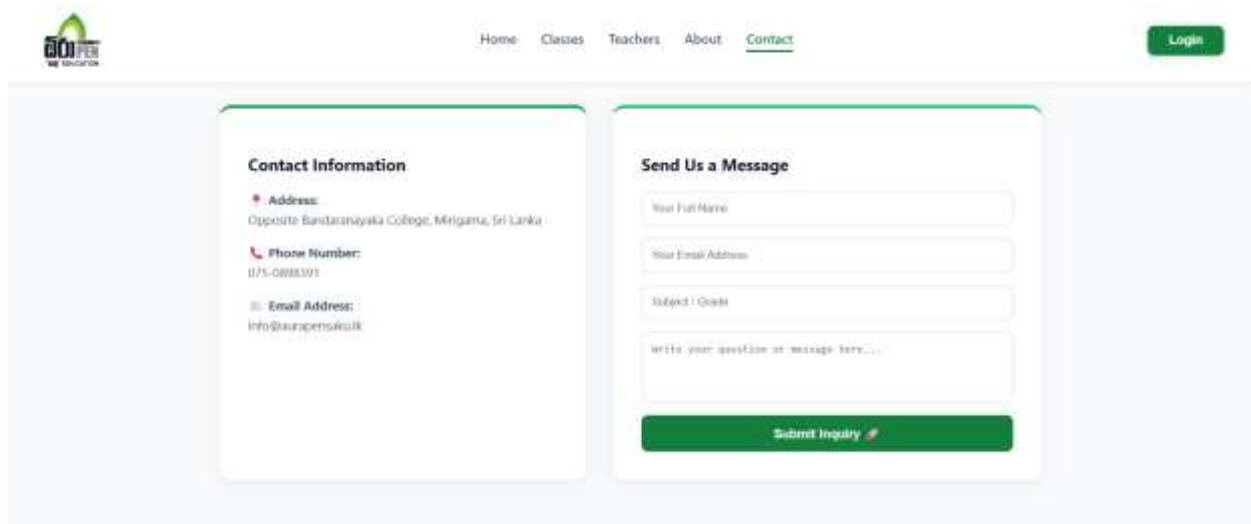


Figure 11 Contact Us Screen

3.5.2 Authentication & Profile Interfaces

1. System Login Interface

Allows Admin, Teacher, and Student users to authenticate using valid credentials and automatically redirects them to their respective role-specific dashboard views.

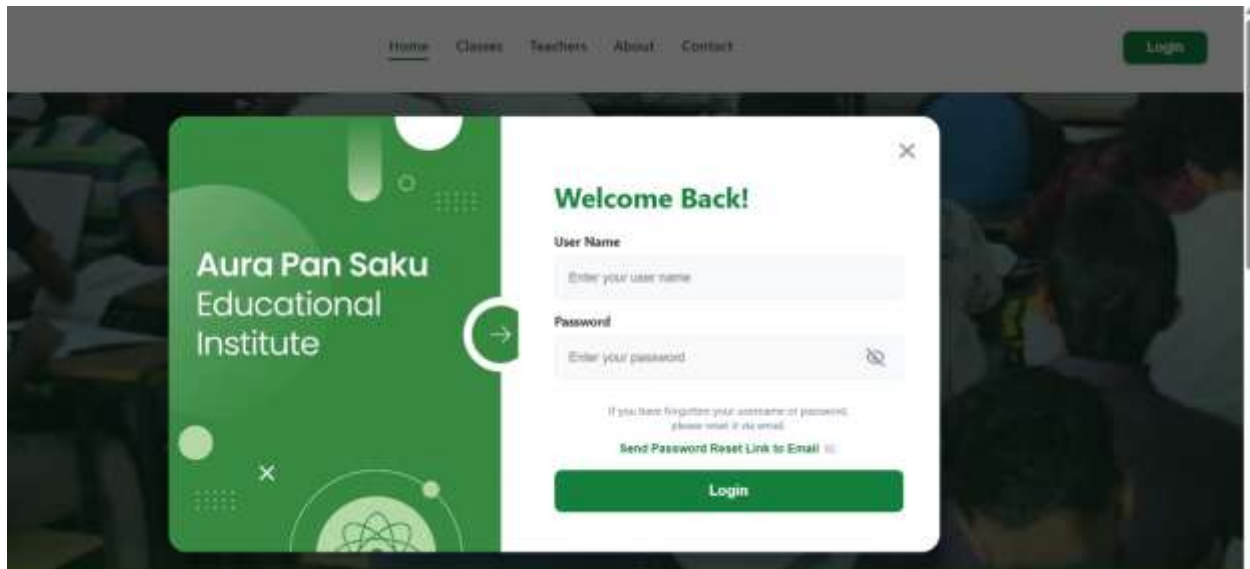


Figure 12 System Login Screen

3.5.3 Admin Portal Interfaces

1. Master Admin Dashboard Overview

Displays a high-level operational overview including total registered students, active teachers, active classes, student enrollment growth trends, grade distribution analytics, and a weekly master timetable matrix.



Figure 13 Admin Dashboard Screen

2. Teacher Management Interface

Enables system administrators to view, search, edit, and manage all faculty member profiles, assigned core subjects, academic qualifications, and contact information.

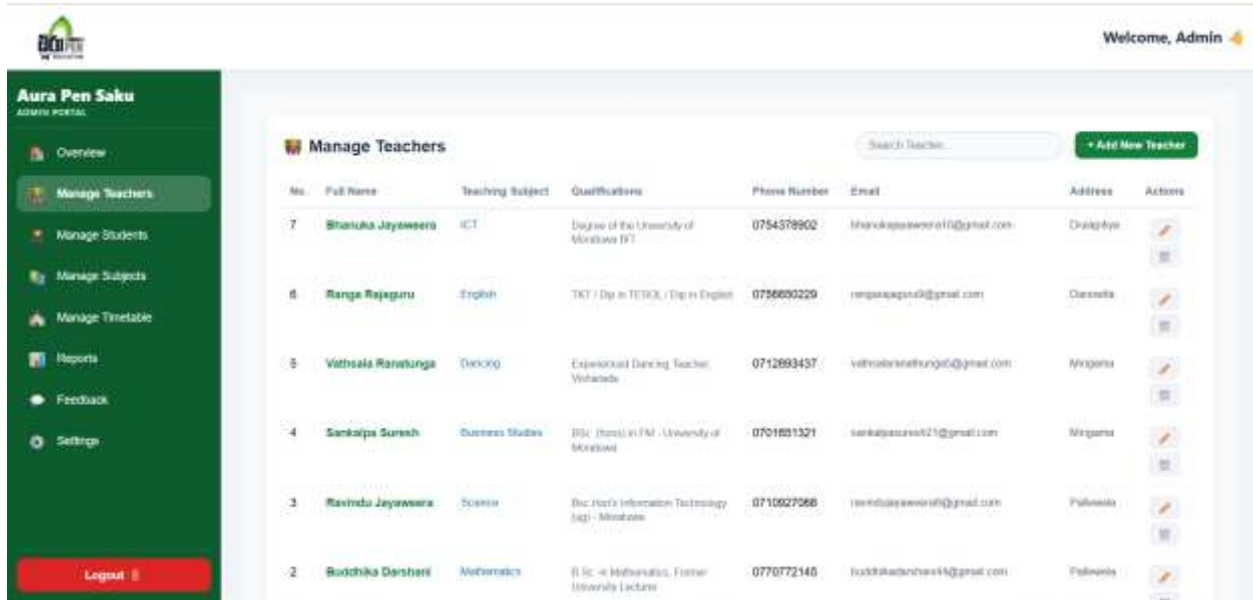


Figure 14 Teacher Management

3. Teacher Registration Modal

An interactive overlay form used by admins to register new teachers by capturing full names, login credentials, assigned teaching subjects, qualifications, and personal contact details.

+ Register New Faculty

Cancel

Save Teacher

Figure 15 Register New Faculty Screen

4. Student Management Interface

Provides a centralized tabular list of all enrolled students, displaying their assigned grade levels, enrolled subject badges, phone numbers, email addresses, physical locations, and profile editing options.

No.	Full Name	Grade	Enrolled Subjects	Phone Number	Email	Address	Actions
7	Lathira Maachanki	Grade 8	Mathematics Science English	9786534386	lathira@maachanki14@gmail.com	Ganegoda	
6	Punali Harshani	Grade 8	Mathematics History Geography	9711321673	punali@harshani33@gmail.com	Ganegoda	
5	Kushila Madushani	Grade 8	Mathematics Science Sinhala	9768612803	kushilam@madushani716@gmail.com	Kilipakula	
4	Anali Perera	Grade 8	Mathematics ICT	9785443286	anali@perera2@gmail.com	Palawala	
3	Rusani Fernando	Grade 8	Mathematics Science English	9711123334	rusani@fernando@gmail.com	Wewalkumbura	
2	Rimal Silva	Grade 8	Business Studies ICT Account	9771894321	rimal@silva12@gmail.com	Muguna	

Figure 16 Manage Students Screen

5. Physical Student Registration Modal

A dedicated popup form allowing admins to onboard new physical students, assign their primary grade level, select multiple course enrollments via checkboxes, and auto-generate login access.

+ Physical Student Registration

Student Full Name

Login Username: ST@123

Grade 10

Select Course Enrollment:

☐ History ☐ Account ☐ Tamil ☐ Sinhala ☐ English ☐ ICT

☐ Dancing ☐ Business Studies ☐ Science ☐ Mathematics

Contact Number (07X...)

Student Email Address

Home Address Location

Cancel Register Student

Figure 17 Student Registration Screen

6. Master Timetable Management Interface

Allows the admin to manage live class schedules across all grade levels (Grade 6 – 12), offering quick filtering by grade and options to edit or delete existing timetable records.

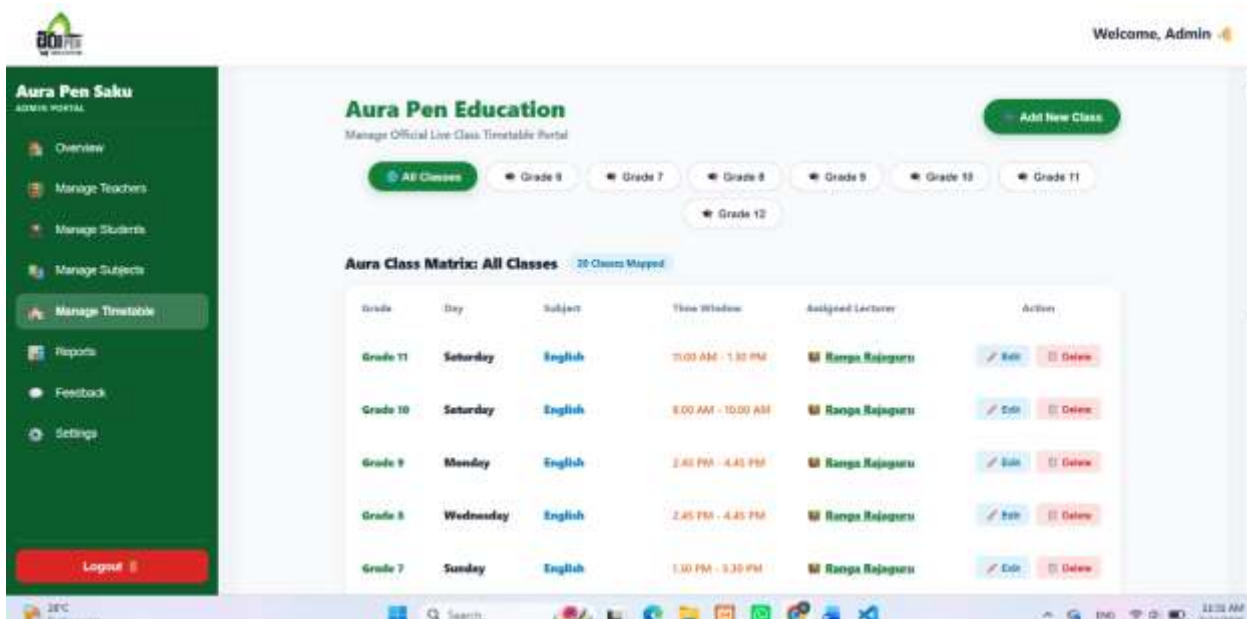


Figure 18 Manage Timetable

7. Class Scheduling Form

An inline management component used to map new live classes by defining target grades, days of the week, time windows, subject names, and assigned lecturers.

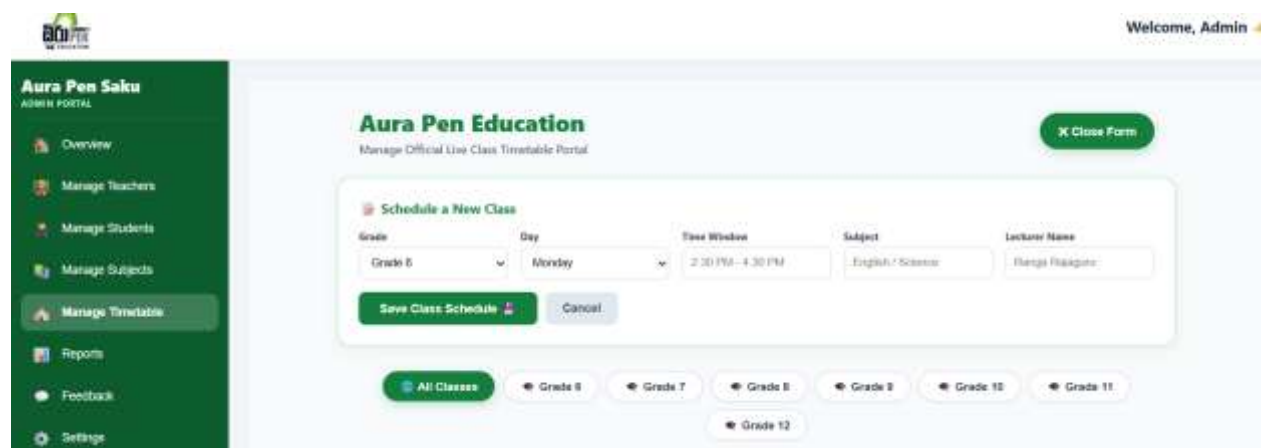


Figure 19 Schedule New Class Screen

8. Academic Performance Reports Dashboard

Provides institutional-level analytics on student examination scores, pass/fail rates, evaluation data matrices, and one-click PDF report generation.

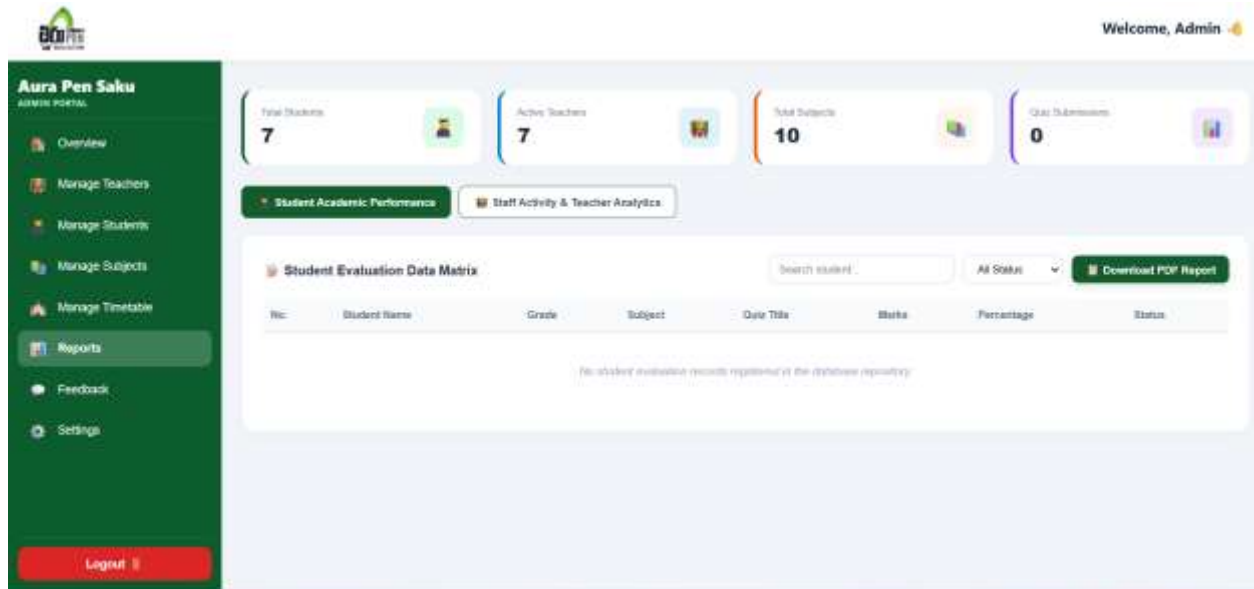


Figure 20 Student Academic Performance Report Screen

9. Staff Operational Analytics & Activity Logs

Tracks operational activity across the teaching staff, displaying active classes managed per lecturer, total published quizzes, and generating downloadable administrative PDF logs.

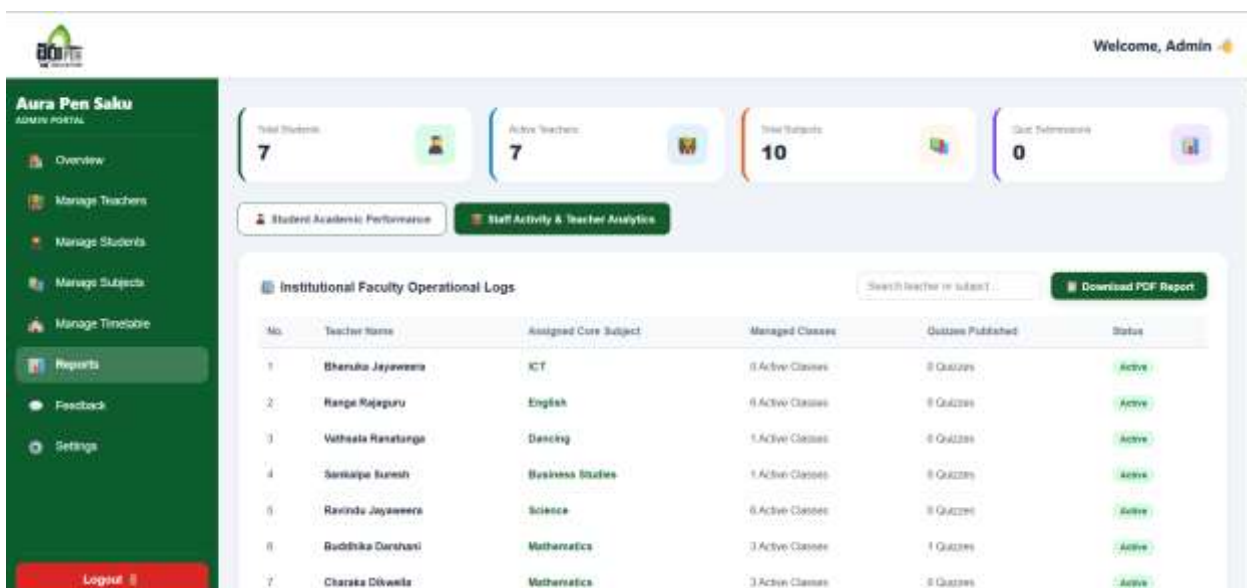


Figure 21 staff Activity & Teacher Analytics Screen

10. Student Feedbacks & App Testimonials Panel

An administrative oversight inbox for reviewing feedback, inquiries, and app ratings submitted by students across various grades and subjects.

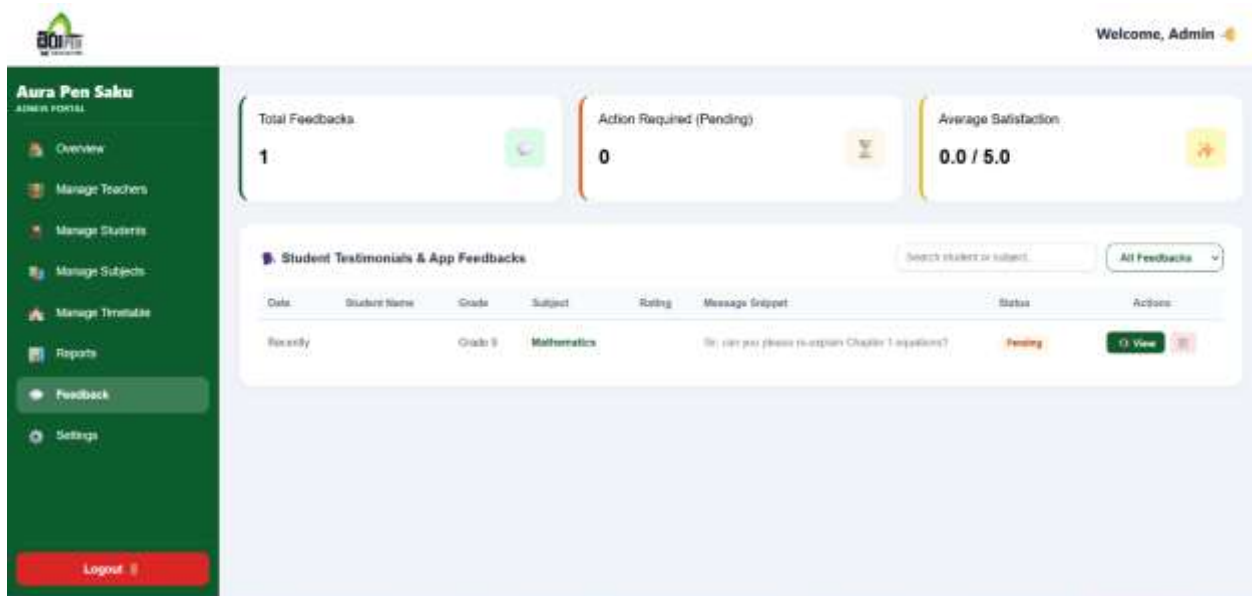


Figure 22 Feedback Management Screen

11. Master System Settings Desk

Allows administrators to update institution branding details (name, official email, helpdesk hotline, physical address), manage admin password security, and trigger app buffer clearing.

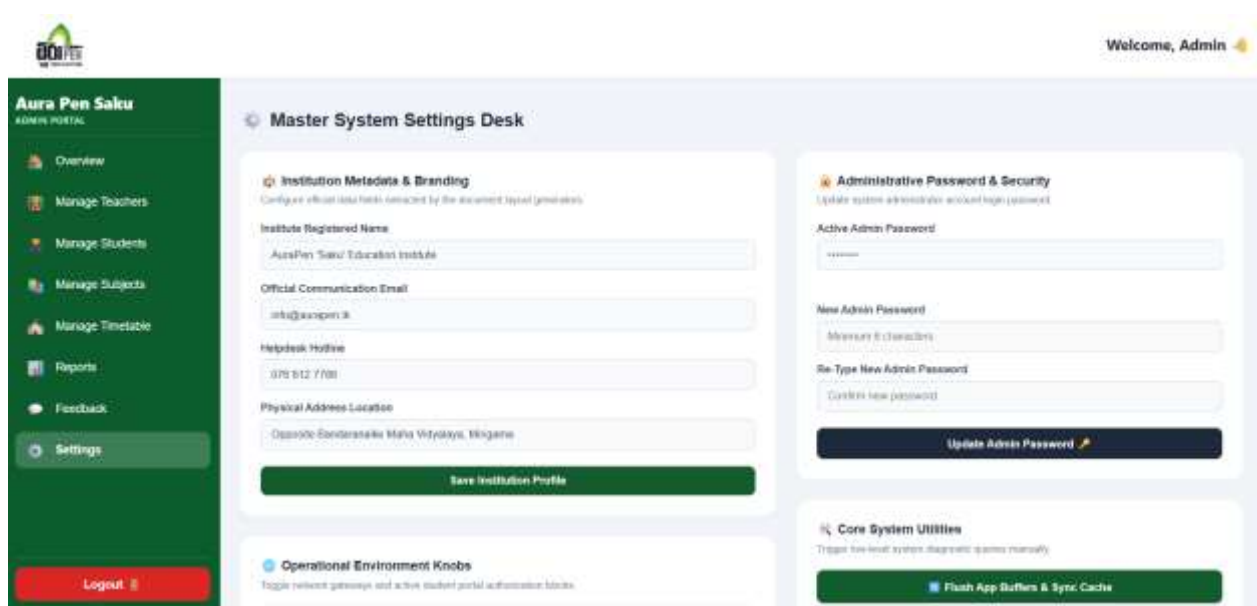


Figure 23 Master System Settings Screen

3.5.4 Teacher Portal Interfaces

1. Teacher Dashboard Overview

Displays a personalized workspace for teachers showing their assigned subjects, managed grade levels, active timetable mapping shortcuts, and quick links to content upload tools.

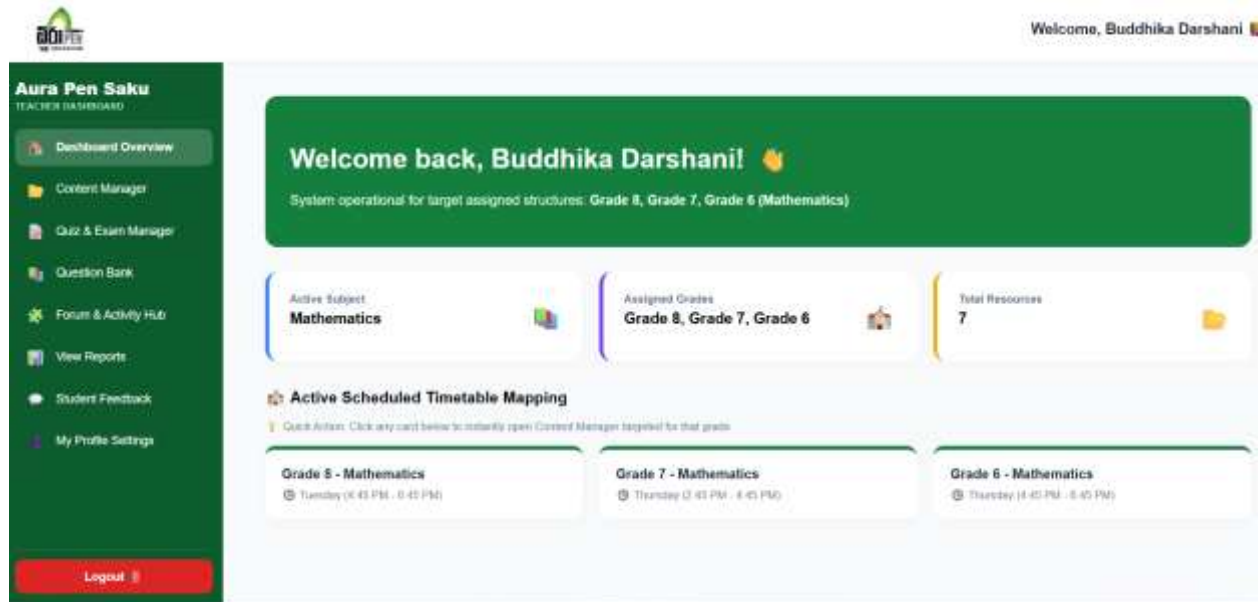


Figure 24 Teacher Dashboard Overview Screen

2. Distributed Content Gateway (Content Manager)

Allows teachers to publish study materials (PDF handouts, video links, handouts) for specific grades and organizes active coursework into lesson-by-lesson modules.

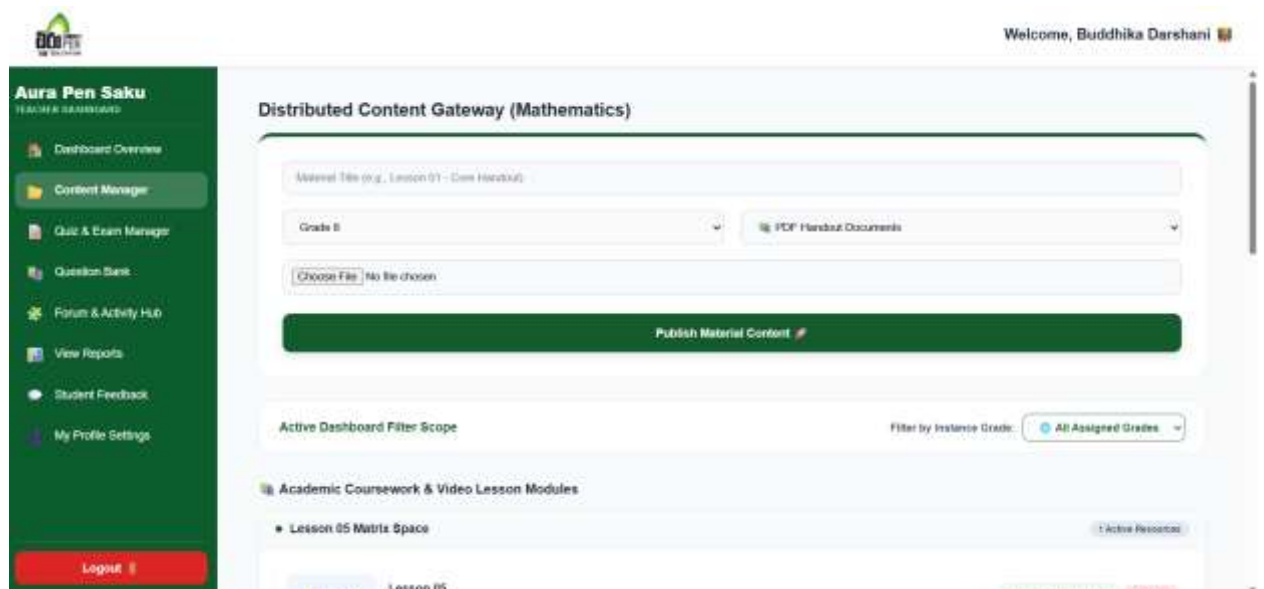


Figure 25 Content Manager Screen

3. Centralized Question Bank Repository

A 3NF relational repository interface where teachers can author master questions (MCQ, Short Answer, True/False), assign target grades, set correct evaluation keys, and filter saved questions by grade and purpose.

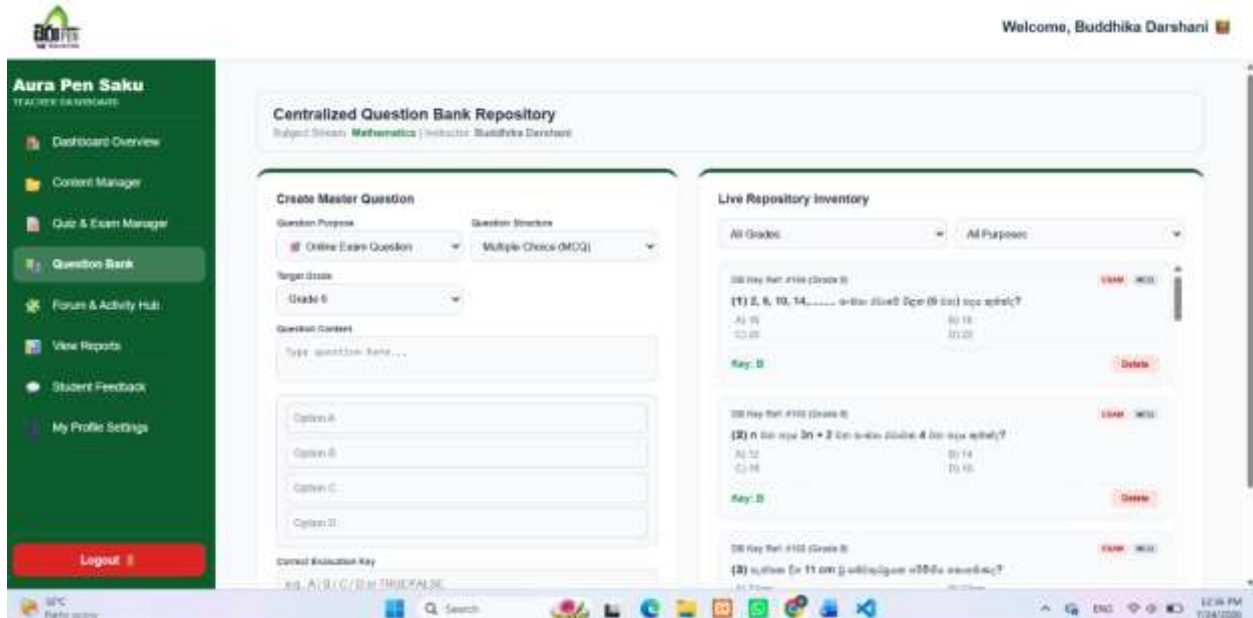


Figure 26 Question Bank Repository Screen

4. Quiz & Exam Manager

An assessment creation tool that allows teachers to build online quiz paper schemas, set duration limits, add custom questions, or link existing questions directly from the central

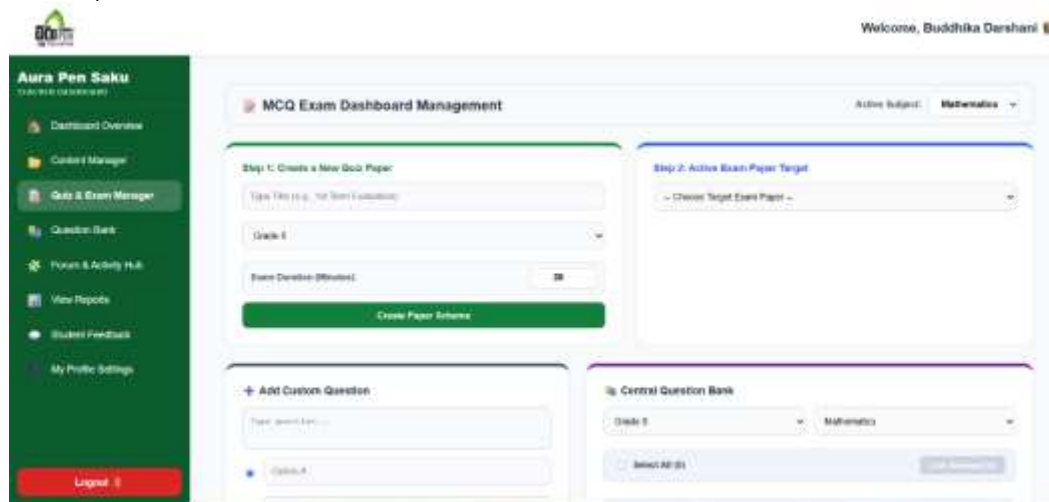


Figure 27 Quiz & Exam Manager Screen

5. Forum & Activity Hub

A dynamic workspace enabling teachers to assign revision activities to specific lesson modules, pulling questions directly from the central bank or creating custom practice items.

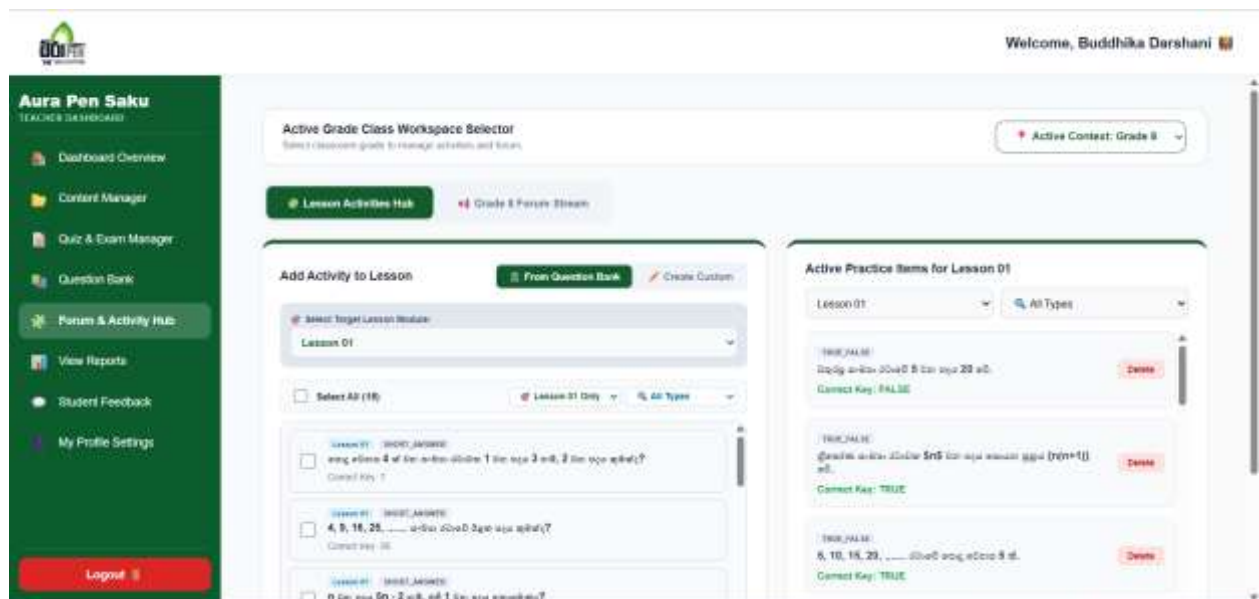


Figure 28 Forum & Activity Hub Screen

6. Quiz & Exam Manager

An assessment creation tool that allows teachers to build online quiz paper schemas, set duration limits, add custom questions, or link existing questions directly from the central repository.



Figure 29 Quiz & Exam Manager Screen

7. Teacher Academic Performance Matrix

Displays real-time analytics on student exam submissions, including class average scores, top scores, pass rates, and individual mark distributions.

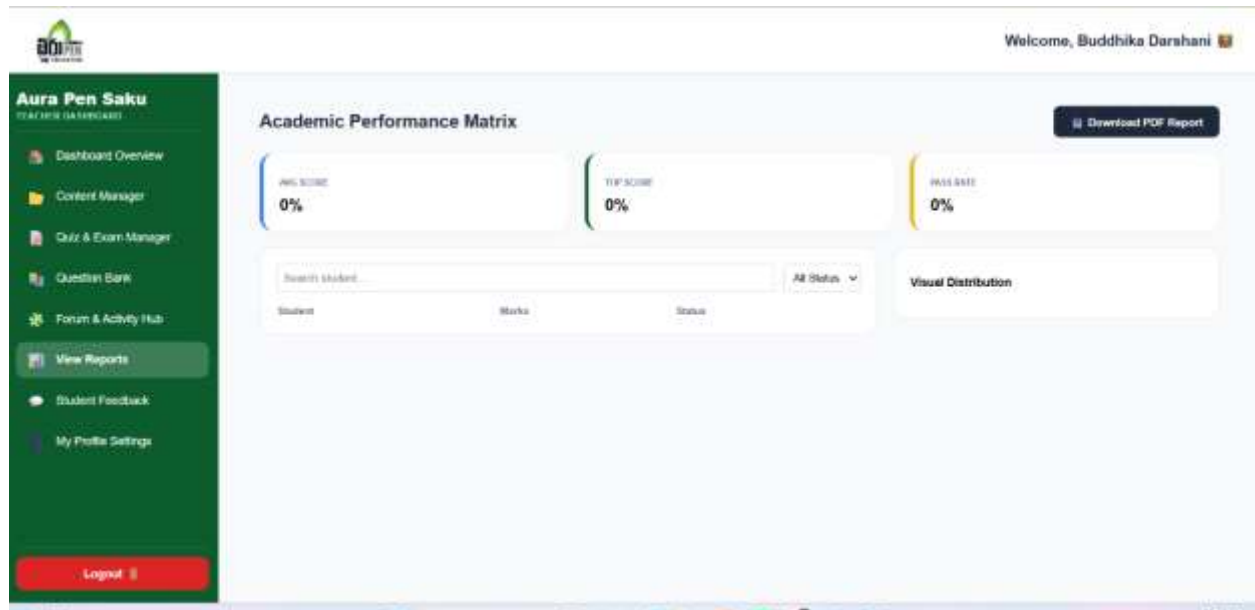


Figure 30 Teacher View Reports Screen

8. Student Feedbacks & Inquiries Inbox

Allows subject teachers to view incoming student queries for their subject and send direct replies to assist students with lesson concepts.

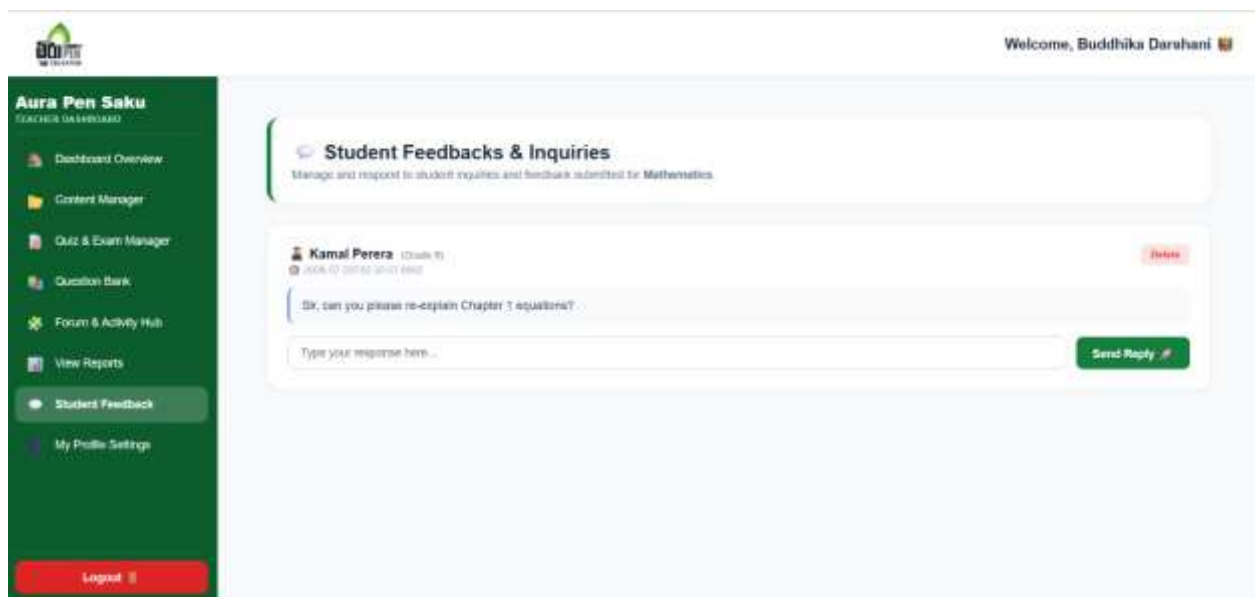


Figure 31 Student Feedback Management Screen

9. Account & Security Settings Panel

Provides faculty members with access to view their registered academic qualifications, assigned grade counts, and update their LMS login username and security password.

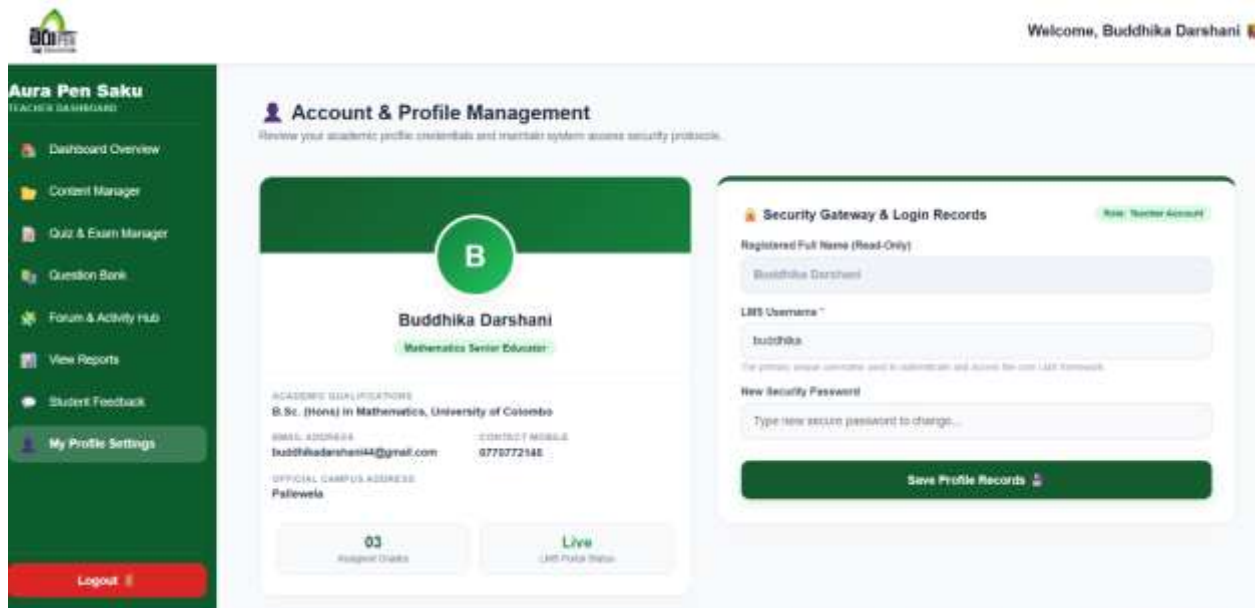


Figure 32 Teacher Profile Settings Screen

3.5.5 Student Portal Interfaces

1. Student Main Dashboard

Displays a personal overview of enrolled subjects, assigned grade status, available study materials, active online exams, live institute announcements, and quick activity shortcuts.

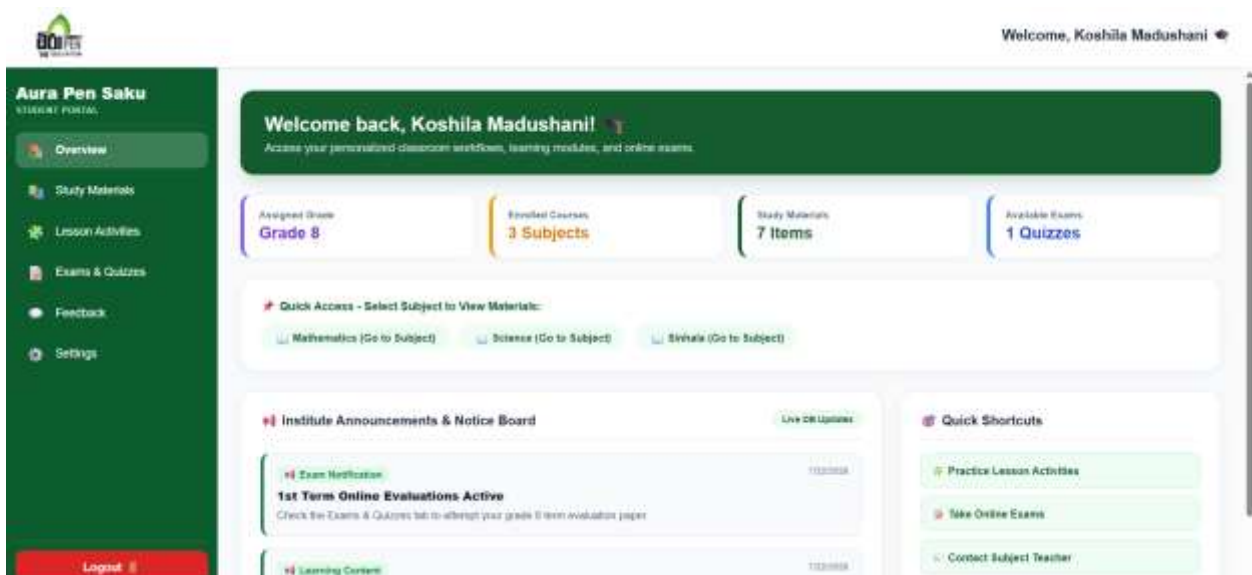


Figure 33 Student Dashboard Screen

2. Course Details & Study Materials Page

Organizes uploaded learning resources (PDF theory notes, exercises, handouts) into structured lesson folders per subject with direct view/download options.

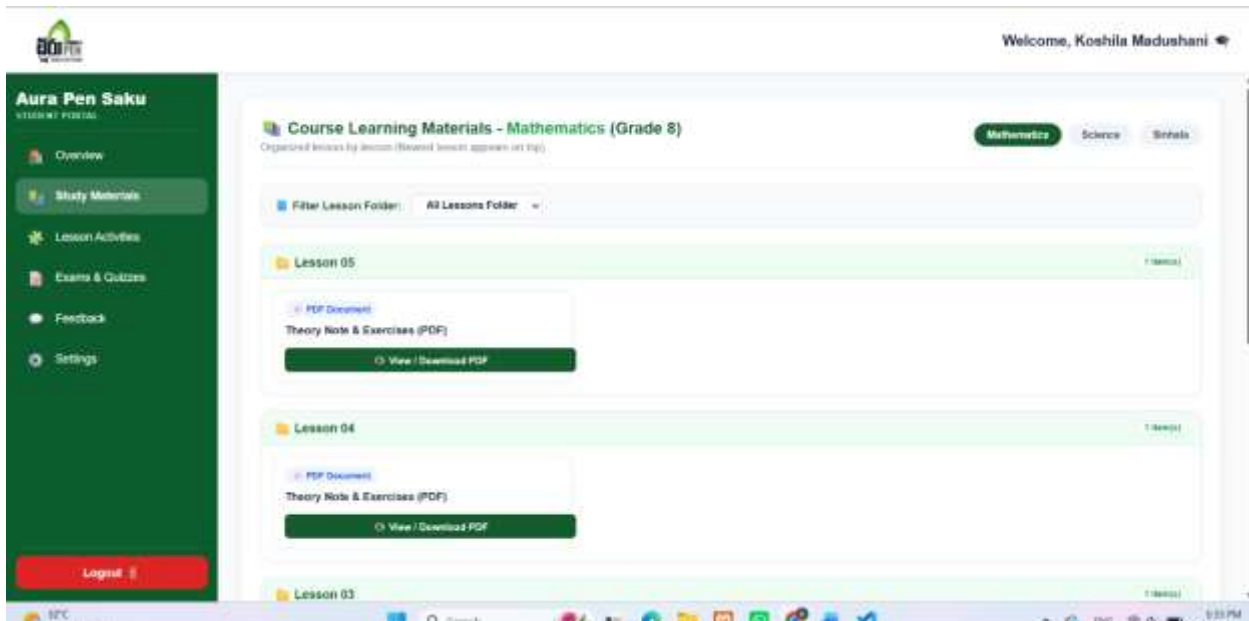


Figure 34 Study Materials Screen

3. Lesson Activities & Practice Hub

Offers interactive practice questions (MCQ, True/False, Short Answer) organized by lesson modules to help students test their understanding of taught concepts.

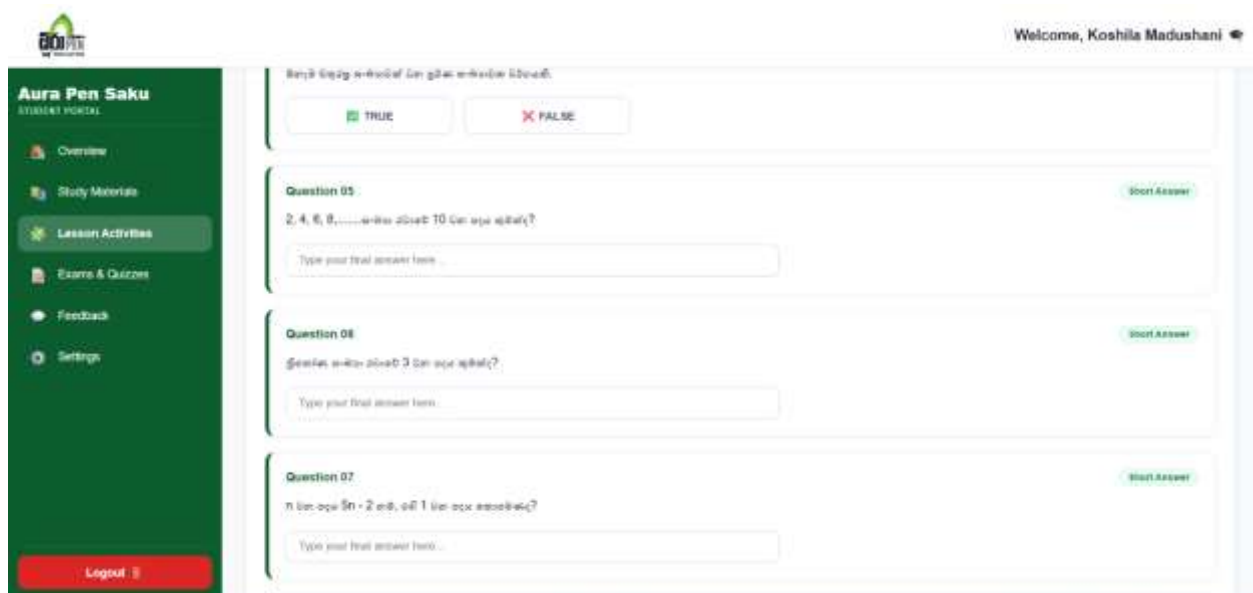


Figure 35 Lesson Activities Screen

4. Online Formal Exams & Evaluation Interface

A secure, timed online examination screen where students attempt multiple-choice questions for published term evaluations and submit answers for instant grading.

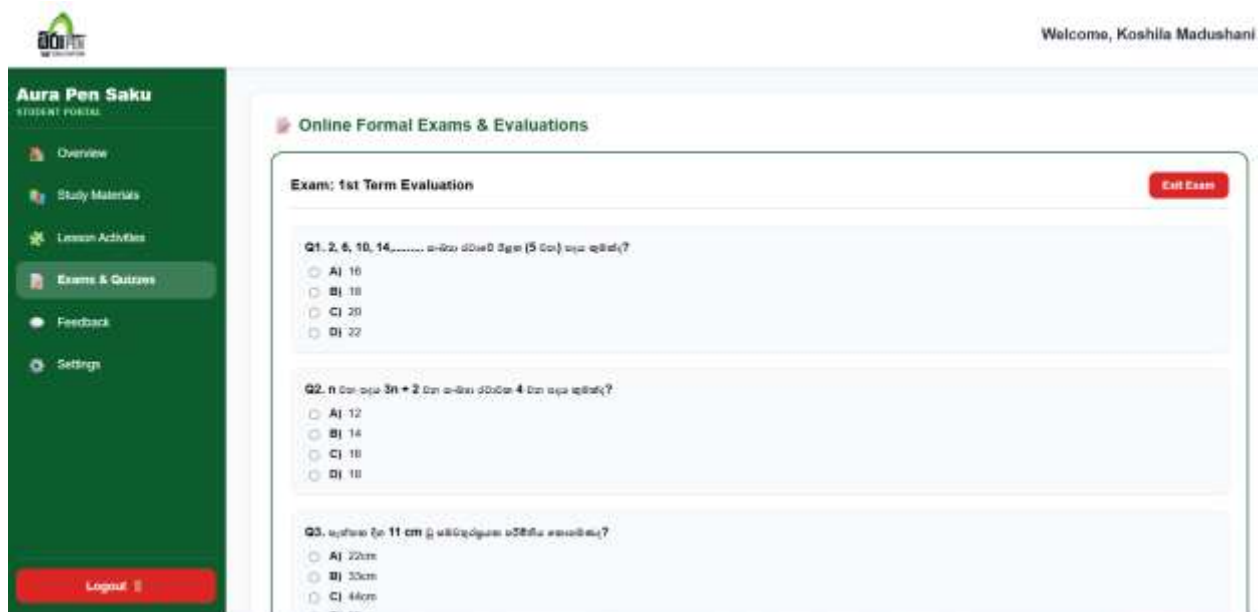


Figure 36 Exams & Quizzes Screen

5. Teacher Feedback & Support Forum

An interactive form allowing students to write and submit academic queries or feedback directly to their subject teachers.

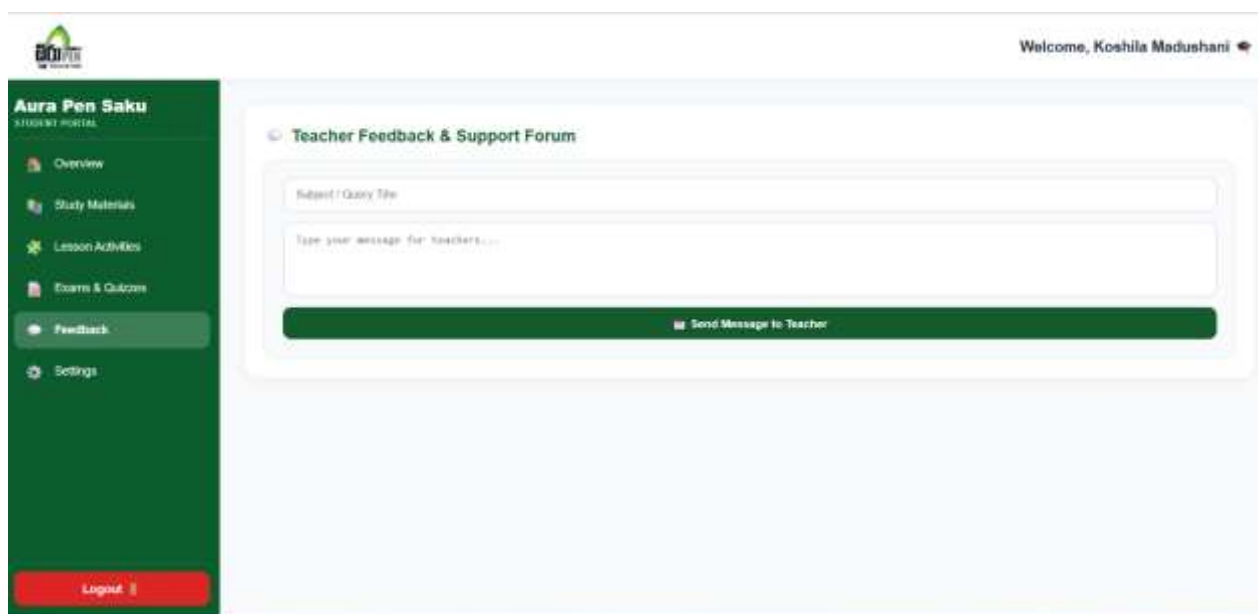


Figure 37 Student Feedback Screen

6. Student Account & Security Settings Page

Displays personal student details (Full Name, Contact Number, Home Address, Enrolled Subjects) and provides a secure panel to update account access passwords.

The screenshot shows the 'Student Profile Settings' page for a user named Koshila Madushani. The page is part of the 'Aura Pen Saku STUDENT PORTAL'. On the left is a green sidebar with navigation links: Overview, Study Materials, Lesson Activities, Exams & Quizzes, Feedback, and Settings (which is highlighted). At the bottom of the sidebar is a red 'Logout' button. The main content area has a header with the user's name 'Koshila Madushani', a profile icon with the letter 'K', and account details: 'Student Account Management & Preferences', 'Grade 6', and 'ID: STU-888'. Below this, there are two main sections: 'Personal Details' and 'Security & Password'. The 'Personal Details' section includes fields for 'Full Name' (Koshila Madushani), 'Contact Phone Number' (0700012000), and 'Home Address' (Kithulala). It also shows 'Enrolled Subjects' as Mathematics, Science, and Sinhala. The 'Security & Password' section includes fields for 'New Password' (Enter new password (min 6 characters)) and 'Confirm New Password' (Re-type new password). A 'Security Tip' box advises to create a password that is hard to guess and to change it regularly. At the bottom right of the main content area is a green 'Save Profile Settings' button.

Welcome, Koshila Madushani

Aura Pen Saku
STUDENT PORTAL

Overview
Study Materials
Lesson Activities
Exams & Quizzes
Feedback
Settings

Logout

Koshila Madushani
Student Account Management & Preferences
Grade 6 ID: STU-888

Personal Details
Update your personal information

Full Name:
Koshila Madushani

Contact Phone Number:
0700012000

Home Address:
Kithulala

Enrolled Subjects:
Mathematics Science Sinhala

Security & Password
Manage your account login credentials

New Password:
Enter new password (min 6 characters)

Confirm New Password:
Re-type new password

Security Tip:
Create password that's hard to guess. If you do not want to change your current account password.

Save Profile Settings

Figure 38 Student Profile Settings Screen

CHAPTER 4 – SYSTEM IMPLEMENTATION

4.1 Introduction to Implementation

This chapter details the technical implementation and architectural deployment of the **AuraPen LMS** web application. The system follows a standard **Three-Tier Architecture** consisting of a React.js presentation tier, a Node.js/Express.js application tier, and a MySQL relational database tier.

4.1.1 Language Used

The system was implemented primarily using **JavaScript (ES6+)** as a unified language across both the frontend (React.js) and backend (Node.js) environments. Structured Query Language (SQL) was utilized for creating schema definitions, foreign key relationships, and data queries against the MySQL database engine.

4.2 Implementation Environment

The development and local deployment environment details are summarized in Table 4.1.

Table 4.1: Implementation Environment

Category	Details
Operating System	Windows 11 Pro (64-bit)
Processor & RAM	13th Gen Intel(R) Core(TM) i5-13420H (2.10 GHz) 16.0 GB RAM
Frontend Technologies	React.js (v18/v19), React Router (Client Routing), Axios (HTTP Client)
Backend Technologies	Node.js, Express.js Framework, Multer (File Upload Middleware), CORS
Database	MySQL 8.0 (accessed via XAMPP Control Panel & mysql2 driver)
IDE & Version Control	Visual Studio Code, Git, Google Chrome / Microsoft Edge Browsers

Table 4:1 Implementation Environment

4.3 Reusable Components

To promote modularity and reduce redundancy, the system was structured into key reusable React components:

- **Navigation Sidebar / Header:** A single reusable sidebar menu component that dynamically renders navigation links based on the authenticated user's role (Admin, Teacher, or Student).
- **Dashboard Overview Cards:** Modular metric widgets used across portals to display aggregated counters (e.g., total enrolled students, uploaded materials, active notices).
- **Chart Visualization Components:** Reusable reporting blocks (powered by Chart.js/Recharts) utilized on performance screens.
- **Modal Dialog Components:** Reusable pop-up modals for password resets, student additions, and schedule editing.
- **Shared Material Component Pattern:** A unified file list and upload pattern reused between Teacher content management and Student download views.

CHAPTER 5- EVALUATION & TESTING

5.1 System Testing Test Plan

The main objective of system testing for the **Aura Pen Saku LMS** is to verify that all functional and non-functional requirements operate as intended without errors. Testing ensures system stability, data security, and seamless user interaction across different roles (Admin, Teacher, and Student).

- **Testing Types Included:** Unit Testing, Integration Testing, System Testing, and User Acceptance Testing (UAT).
- **Testing Environment:** Windows 11, Google Chrome Browser, Node.js / Express.js Backend, MySQL (phpMyAdmin - aurapen_lms_db) Database, React.js Frontend.
- **Pass/Fail Criteria:** A test case is marked as **PASS** if the *Actual Result* matches the *Expected Result*. Any deviation marks the test case as **FAIL**.

5.2 Detailed System Test Cases & Results

Test Case ID: TC_01 - Valid System Login

Field	Details
Module / Entity	User Authentication (users table)
Test Description	Verify that a registered user can successfully log in with valid credentials.
Pre-conditions	User account exists in the database with active status.
Test Steps	1. Navigate to LMS Login Page. 2. Enter valid username and password. 3. Click 'Login' button.
Expected Result	System authenticates user and redirects to the role-based dashboard.
Actual Result	User authenticated successfully and redirected to dashboard.
Status	PASS

Table 5-1:TC_01 - Valid System Login

Test Case ID: TC_02 - Invalid Password Handling

Field	Details
Module / Entity	User Authentication (users table)
Test Description	Verify that login is rejected when an incorrect password is entered.
Pre-conditions	User account exists in the database.
Test Steps	1. Enter valid username. 2. Enter incorrect password. 3. Click 'Login' button.
Expected Result	Access denied and error message displayed: <i>"Invalid credentials"</i> .
Actual Result	Error message displayed as expected. Access denied.
Status	PASS

*Table 5-2:TC_02 – Invalid Password***Test Case ID: TC_03 - Empty Input Validation**

Field	Details
Module / Entity	User Authentication (users table)
Test Description	Verify that the system prevents submission when login fields are left blank.
Pre-conditions	LMS Login Page loaded.
Test Steps	1. Leave username and password fields blank. 2. Click 'Login' button.
Expected Result	Form submission blocked and prompt displayed: <i>"Fields cannot be empty"</i> .
Actual Result	Validation alert displayed. Form submission blocked.
Status	PASS

Table 5-3:TC_03 – Empty Input Validation

Test Case ID: TC_04 - First-Time Password Reset

Field	Details
Module / Entity	User Security (is_first_login flag)
Test Description	Verify that first-time login forces the user to reset their temporary password.
Pre-conditions	User account has is_first_login = TRUE in DB.
Test Steps	1. Log in using temporary password. 2. Enter new password in reset screen. 3. Click 'Update Password'.
Expected Result	Password updated and is_first_login flag updated to FALSE in database.
Actual Result	Forced reset prompted, password updated, and flag set to FALSE.
Status	PASS

*Table 5-4:TC_04 – first Time Password Reset***Test Case ID: TC_05 - Add Teacher Profile**

Field	Details
Module / Entity	Teacher Management (teachers table)
Test Description	Verify Admin can create a new teacher profile linked to a user account.
Pre-conditions	Admin logged into system; user account created for teacher.
Test Steps	1. Navigate to Teacher Management. 2. Input teacher full name, email, phone, and user_id. 3. Click 'Save'.
Expected Result	Record stored in teachers table with Foreign Key linking to users table.
Actual Result	Teacher record created and successfully linked.
Status	PASS

Table 5-5:TC_5 – Add Teacher Table

Test Case ID: TC_06 - Duplicate Email Prevention

Field	Details
Module / Entity	Data Validation (teachers / students tables)
Test Description	Verify that duplicate email registration is rejected by UNIQUE constraint.
Pre-conditions	Teacher/Student record with target email already exists.
Test Steps	1. Input existing registered email. 2. Click 'Save'.
Expected Result	Database rejects insertion and displays: <i>"Email already exists"</i> .
Actual Result	Rejection message displayed correctly. Duplicate prevented.
Status	PASS

*Table 5-6:TC_06 – Duplicate email***Test Case ID: TC_07 - Student Profile Update**

Field	Details
Module / Entity	Student Portal (students table)
Test Description	Verify student can update their contact phone number and address.
Pre-conditions	Student logged into Student Portal.
Test Steps	1. Open 'My Profile'. 2. Update phone number and address. 3. Click 'Save Changes'.
Expected Result	Student record updated in DB and changes reflected on screen.
Actual Result	Profile updated successfully in DB and UI.
Status	PASS

Table 5-7:TC_07 – Student Profile

Test Case ID: TC_08 - Create New Subject

Field	Details
Module / Entity	Subject Management (subjects table)
Test Description	Verify Admin can register a new subject with unique code and title.
Pre-conditions	Admin logged into system.
Test Steps	1. Go to Subject Management. 2. Input subject_code and subject_name. 3. Click 'Add Subject'.
Expected Result	Subject created in subjects table enforcing UNIQUE constraint on subject_code.
Actual Result	Subject inserted into database successfully.
Status	PASS

*Table 5-8:TC_08 – Create New Subject***Test Case ID: TC_09 - Duplicate Subject Code Check**

Field	Details
Module / Entity	Subject Management (subjects table)
Test Description	Verify that creating a subject with an existing subject code is blocked.
Pre-conditions	Subject code already exists in DB.
Test Steps	1. Enter duplicate subject_code. 2. Click 'Save'.
Expected Result	System blocks submission and displays error alert.
Actual Result	Error alert displayed. Submission blocked.
Status	PASS

Table 5-9:TC_09 – Duplicate Subject

Test Case ID: TC_10 - Assign Subject to Teacher

Field	Details
Module / Entity	Course Mapping (teacher_subjects table)
Test Description	Verify Admin can assign a teacher to a specific subject module.
Pre-conditions	Teacher and Subject exist in DB.
Test Steps	1. Select teacher_id and subject_id. 2. Click 'Assign Subject'.
Expected Result	Mapping record stored in teacher_subjects table.
Actual Result	Mapping record saved; subject displayed in teacher portal.
Status	PASS

*Table 5-10:TC_10 – Assign Subject***Test Case ID: TC_11 - Student Course Enrollment**

Field	Details
Module / Entity	Student Mapping (student_subjects table)
Test Description	Verify student can enroll in an available subject module.
Pre-conditions	Student account active; subject available.
Test Steps	1. Select target subject. 2. Click 'Enroll Now'.
Expected Result	Enrollment record created in student_subjects table.
Actual Result	Enrolled successfully; subject listed in student dashboard.
Status	PASS

Table 5-11:TC_11 – student Course

Test Case ID: TC_12 - Valid PDF Material Upload

Field	Details
Module / Entity	Content Management (learning_materials table)
Test Description	Verify teacher can upload lecture notes in PDF format.
Pre-conditions	Teacher assigned to subject.
Test Steps	1. Select subject and attach valid PDF file. 2. Enter material title. 3. Click 'Upload'.
Expected Result	File saved to server path and file_path stored in learning_materials table.
Actual Result	PDF uploaded and visible under student study materials.
Status	PASS

*Table 5-12:TC_12- Valid Pdf***Test Case ID: TC_13 - Invalid File Extension Upload**

Field	Details
Module / Entity	Upload Validation (learning_materials table)
Test Description	Verify system blocks unsupported file extensions (e.g., .exe, .mp3).
Pre-conditions	Teacher on Upload page.
Test Steps	1. Attach an executable file (.exe). 2. Click 'Upload'.
Expected Result	Upload rejected and error shown: <i>"Invalid file format. Upload PDF/DOCX"</i> .
Actual Result	Upload blocked and error prompt displayed.
Status	PASS

Table 5-13:TC_13 – Invalid File

Test Case ID: TC_14 - Oversized File Upload

Field	Details
Module / Entity	Upload Validation (learning_materials table)
Test Description	Verify system blocks files exceeding maximum upload size limit (50MB).
Pre-conditions	Teacher on Upload page.
Test Steps	1. Attach PDF file > 50MB. 2. Click 'Upload'.
Expected Result	System blocks upload and displays max file size notice.
Actual Result	Error message displayed. Upload blocked.
Status	PASS

*Table 5-14:TC_14 – Oversized File***Test Case ID: TC_15 - Create Lesson Activity**

Field	Details
Module / Entity	Lesson Management (lesson_activities table)
Test Description	Verify teacher can post a new lesson topic or activity.
Pre-conditions	Subject module selected.
Test Steps	1. Enter activity_title and description. 2. Click 'Publish'.
Expected Result	Activity record created in lesson_activities table.
Actual Result	Activity created and viewable by students under lesson view.
Status	PASS

Table 5-15:TC_15 – Create Lesson

Test Case ID: TC_16 - Create Online Quiz

Field	Details
Module / Entity	Quiz Management (quizzes table)
Test Description	Verify teacher can create an online quiz for a subject.
Pre-conditions	Teacher logged into system.
Test Steps	1. Enter quiz_title and select subject_id. 2. Click 'Save Quiz'.
Expected Result	Quiz metadata inserted into quizzes table.
Actual Result	Quiz created and ready for question assignment.
Status	PASS

*Table 5-16:TC_16 – Create Online Quiz***Test Case ID: TC_17 - Add MCQ Questions**

Field	Details
Module / Entity	Quiz Questions (questions table)
Test Description	Verify teacher can add MCQ questions with options to a quiz.
Pre-conditions	Quiz created in system.
Test Steps	1. Input question_text and 4 options. 2. Select correct_answer. 3. Click 'Add Question'.
Expected Result	Question record saved in questions table linked to quiz_id.
Actual Result	Question attached to quiz successfully.
Status	PASS

Table 5-17:TC_17 – Add MCQ Question

Test Case ID: TC_18 - Master Question Bank Sync

Field	Details
Module / Entity	Question Repository (question_bank table)
Test Description	Verify questions can be saved into global question bank for future reuse.
Pre-conditions	Question created.
Test Steps	1. Check 'Save to Question Bank'. 2. Click 'Submit'.
Expected Result	Question record inserted into question_bank table under subject_id.
Actual Result	Record saved in question bank table successfully.
Status	PASS

*Table 5-18:TC_18 – Master Question***Test Case ID: TC_19 - Automated Quiz Grading**

Field	Details
Module / Entity	Quiz Assessment (student_results table)
Test Description	Verify system automatically calculates and saves student quiz marks upon submission.
Pre-conditions	Student actively attempting quiz.
Test Steps	1. Select answers for MCQ questions. 2. Click 'Submit Quiz'. 3. Confirm submission.
Expected Result	Score evaluated, record saved in student_results table, and final score shown.
Actual Result	Score calculated instantly and saved in DB.
Status	PASS

Table 5-19:TC_19 – Create Online Quiz

Test Case ID: TC_20 - Quiz Timer Auto-Submission

Field	Details
Module / Entity	Quiz Execution (quizzes table)
Test Description	Verify system auto-submits answered questions when timer expires.
Pre-conditions	Timed quiz started by student.
Test Steps	1. Answer partial questions. 2. Allow timer to reach 00:00 without clicking submit.
Expected Result	System auto-submits answers, calculates score, and saves result.
Actual Result	Auto-submitted on timer expiration and score recorded.
Status	PASS

*Table 5-20:TC_20 – Quiz Timer***Test Case ID: TC_21 - Post Announcement**

Field	Details
Module / Entity	Notice Board (announcements table)
Test Description	Verify Admin/Teacher can post notices visible on student feeds.
Pre-conditions	Admin/Teacher logged into portal.
Test Steps	1. Enter title and content message. 2. Click 'Post Announcement'.
Expected Result	Notice saved in announcements table and displayed on student notice board.
Actual Result	Notice published and visible on student dashboard.
Status	PASS

Table 5-21:TC_21 – Post Announcement

Test Case ID: TC_22 - Submit Course Feedback

Field	Details
Module / Entity	Feedback System (feedbacks table)
Test Description	Verify student can submit feedback for a course module.
Pre-conditions	Student enrolled in subject.
Test Steps	1. Enter feedback comment. 2. Click 'Send Feedback'.
Expected Result	Feedback record stored in feedbacks table linked to student_id.
Actual Result	Feedback saved in DB successfully.
Status	PASS

*Table 5-22:TC_22 – Submit Course Feedback***Test Case ID: TC_23 - Add Timetable Slot**

Field	Details
Module / Entity	Live Schedule (timetable table)
Test Description	Verify Admin can add class schedule slots to the live matrix grid.
Pre-conditions	Admin logged into system.
Test Steps	1. Select subject_id, day_of_week, start_time, and end_time. 2. Click 'Save Schedule'.
Expected Result	Entry saved in timetable table and reflected on live matrix calendar.
Actual Result	Timetable updated real-time across public and student portals.
Status	PASS

Table 5-23:TC_23 – Add Time Slot

Test Case ID: TC_24 - Foreign Key Referential Integrity

Field	Details
Module / Entity	Database Engine (MySQL Constraints)
Test Description	Verify deleting a parent user record handles linked student/teacher records.
Pre-conditions	User record linked to student profile.
Test Steps	1. Admin deletes user from users table.
Expected Result	Database CASCADE/RESTRICT enforces referential integrity without orphan records.
Actual Result	Referential integrity enforced as configured in database schema.
Status	PASS

Table 5-24:TC_24 – Foreign Key Referential Integrity

Test Case ID: TC_25 - Role Access Security

Field	Details
Module / Entity	System Security (RBAC Middleware)
Test Description	Verify unauthorized direct URL access to Admin Portal by student is blocked.
Pre-conditions	Student logged into system.
Test Steps	1. Paste Admin Portal URL directly into browser address bar. 2. Press Enter.
Expected Result	System blocks access and redirects student to login or unauthorized page.
Actual Result	Access blocked; redirected to unauthorized access page.
Status	PASS

Table 5-25:TC_25 – Role Access Security

5.3 User Acceptance Test (UAT)

User Acceptance Testing was conducted to evaluate system usability, operational readiness, and satisfaction among end-users. The UAT panel comprised **3 Teachers, 10 Students, and 1 System Administrator**.

UAT Check List & User Agreement

- **User Agreement:** Participants evaluated system functionalities using realistic academic data under standard operating conditions.
- **Evaluation Checklist:**
 1. Ease of navigation and interface clarity.
 2. Speed and stability of learning material downloads.
 3. Accuracy of automated quiz evaluations.
 4. Matrix timetable responsiveness across mobile and desktop devices.

5.4 User Feedback

- **Teacher Feedback:** Teachers appreciated the simplified workflow for uploading PDF notes and generating online quizzes. They suggested adding bulk enrollment options in future iterations.
- **Student Feedback:** Students highlighted the convenience of instant score generation for quizzes and real-time announcements on their dashboard.
- **Admin Feedback:** Administrators confirmed that role-based access controls (RBAC) and matrix schedule management worked smoothly without system delays.

5.5 User Acceptance Test Results

Feature Evaluated	Evaluation Criteria	Satisfactory Rating (%)	Status
User Authentication	Smooth login, password reset, and role redirect	96%	Accepted
Course & Material Upload	Easy PDF upload, file preview, and deletion	92%	Accepted
Quiz Execution	Timer accuracy, instant score calculation	95%	Accepted
User Interface & UI	Navigation ease and screen responsiveness	90%	Accepted
Overall System Stability	No critical system crashes or data loss during tests	98%	Accepted

Table 5-26– User Acceptance Test Result

CHAPTER 6 – CONCLUSION & FUTURE ENHANCEMENTS

6.1 Introduction

The **AuraPen LMS** project was designed and developed to automate and digitize the academic and operational workflows at Aura Pen Saku Education Institute. By replacing manual paper registers, physical tute printing, and verbal announcements with a role-based web application, the system streamlines student management, study material distribution, timetable tracking, and online quiz evaluations into a single platform. Overall, the project successfully achieved all specified project aims and functional objectives.

6.2 Lessons Learnt

Developing this full-stack web application provided valuable practical experience across modern software frameworks, libraries, and system architecture design:

- **Frontend Development (React.js):** Gained hands-on experience in building dynamic single-page application (SPA) interfaces, component-based architecture, state management (useState, useEffect), and handling REST API calls using Axios.
- **Backend Development (Node.js & Express.js):** Learned to design RESTful API endpoints, handle asynchronous middleware requests, implement Multer for physical file upload handling, and secure protected routes.
- **Database Management (MySQL):** Practical knowledge gained in designing normalized relational schemas, establishing foreign key constraints across tables, and running efficient SQL queries.
- **Framework Selection & Integration:** Understood how to integrate different open-source frameworks and libraries cohesively into a three-tier software architecture.

6.3 Future Enhancements

While the current web system completely satisfies the institute's core requirements, the following features are planned for future versions:

- **Parent SMS / WhatsApp Notifications:** Implementing an automated SMS gateway to send instant attendance and exam result alerts directly to parents' mobile phones.
- **Native Mobile Application:** Developing standalone native mobile apps for Android (.apk) and iOS (.ipa) using React Native for a smoother mobile experience.
- **Online Payment Gateway:** Integrating payment portals (e.g., PayHere or Credit Card gateways) to allow online class fee payments and digital receipt generation.
- **Automated Certificate Generation:** Adding a feature to generate downloadable course completion certificates for students automatically upon quiz completion.

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APPENDIX A – REQUIREMENT GATHERING

A.1 Client Interview Summary

Category	Details
Project Title	Web-Based Learning Management System (LMS) for Aura Pen Saku Education Institute
Interviewee	Mr. Sameera Kaluarachchi (Director / Client)
Interviewer	P.P.K.M. Gunathilaka
Date & Venue	Aura Pen Saku Education Institute, Mirigama
Initial System Status	The institute operated entirely using manual, paper-based administrative methods prior to this project.
Key Operational Issues	• Managing dynamic timetable modifications manually. • Distributing printed lesson tutes physically. • Maintaining paper registers for student records.
Client Requirements	• An easy-to-use dynamic timetable matrix view. • PDF tute upload facility grouped by grade and subject. • Dynamic notice cards for live announcements.

Table A-1 – Client Interview Record & Findings

A.2 User Questionnaire Sample & Survey Summary

Below is the structured requirement gathering survey issued to students and teaching staff during the initial analysis phase.

Question No.	Survey Question / Evaluation Item	Response Options	Majority Stakeholder Response
Q1	How are class timetables and schedule modifications currently communicated to students?	☐ Physical Notice Board ☐ Printed Papers / Handouts ☐ Verbal Class Announcements	Verbal Class Announcements & Whiteboard
Q2	Would digital PDF downloads for tutes and lesson notes reduce printing costs and save distribution time?	☐ Strongly Agree ☐ Agree ☐ Neutral ☐ Disagree	☒ Strongly Agree
Q3	Is an automated online MCQ quiz system preferred over manual paper-based test marking?	☐ Strongly Agree ☐ Agree ☐ Neutral ☐ Disagree	☒ Strongly Agree
Q4	Would a centralized web portal improve overall access to institute notices and course updates?	☐ Strongly Agree ☐ Agree ☐ Neutral ☐ Disagree	☒ Strongly Agree

Table A-2: User Requirement Survey & Response Summary

APPENDIX B – SOURCE CODE SNIPPETS & TECHNICAL LOGS

B.1 Database Engine & Schema Verification

- **Database Management System:** MySQL Relational Database Engine (phpMyAdmin)
- **Database Name:** aurapen_lms_db
- **Tables Verified:** 15 Relational Tables (users, teachers, students, subjects, teacher_subjects, student_subjects, learning_materials, lesson_activities, quizzes, questions, question_bank, student_results, announcements, feedbacks, timetable).
- **Relational Integrity:** Foreign Keys defined with CASCADE and RESTRICT actions to ensure zero orphaned records

B.2 System Development & Execution Environment

Layer	Technology Stack	Configuration Details
Front-End Layer	React.js (ES6+), HTML5, CSS3	Single Page Application (SPA) responsive design
Back-End Layer	Node.js / Express.js	RESTful APIs for handling asynchronous HTTP requests
Database Layer	MySQL (aurapen_lms_db)	Relational database executed via phpMyAdmin environment
Deployment Server	Localhost Express Server	Executing on http://localhost:5000

B.2 System Development & Execution Environment